

AI BASED SPOOF DETECTION

A Project Report Submitted in partial fulfilment of the requirements for the award of the degree

of

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Submitted by

2210040007: S.B. Hema Anjali

2210040032: CH. Samyana Reddy

2210040033: P. Nikhita Reddy

2210040044: P. Sree Varsha Reddy

Batch No:

2204004H

Under the guidance of

Dr. A.K. Varma



Department of Electronics and Communication Engineering

Koneru Lakshmaiah Education Foundation, Aziz Nagar

Aziz Nagar – 500075

APRIL- 2026

KONERU LAKSHMAIAH EDUCATION FOUNDATION

Department of Electronics and Communication Engineering



DECLARATION

The Project Report entitled “**AI BASED SPOOF DETECTION**” is a record of Bonafide work of **S. B. Hema Anjali(2210040007)**, **Ch. Samyana Reddy (2210040032)**, **P. Nikhita Reddy (2210040033)**, **P. Sree Varsha Reddy (2210040044)** submitted in partial fulfillment for the award of B. Tech in Department of Electronics and Communication Engineering to the Koneru Lakshmaiah Education Foundation (Deemed to be University). The results embodied in this report have not been copied from any other departments/University/Institute.

S. B. Hema Anjali	(2210040007)
CH. Samyana Reddy	(2210040032)
P. Nikhita Reddy	(2210040033)
P. Sree Varsha Reddy	(2210040044)

KONERU LAKSHMAIAH EDUCATION FOUNDATION

Department of Electronics and Communication Engineering



CERTIFICATE

This is to certify that the Project Report entitled “**AI BASED SPOOF DETECTION**” is a record of Bonafide work of **S. B. Hema Anjali (2210040007), Ch. Samyana Reddy (2210040032), P. Nikhita Reddy (2210040033), P. Sree Varsha Reddy (2210040044)** submitted in partial fulfilment for the award of B. Tech in Department of Electronic and communication Engineering to the Koneru Lakshmaiah Education Foundation (Deemed to be University) is a record of bonafide work carried out under my guidance and supervision.

The results embodied in this report have not been copied from any other departments / University / Institute.

Signature of the HOD

Dr. B. Anil Kumar
(Head of the department)

Signature of the Supervisor

Dr. A. K. Varma
(Assistant Professor)

Signature of the Internal Examiner

Signature of the External Examiner

Acknowledgement

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Submitted By

S. B. Hema Anjali	(2210040007)
Ch. Samyana Reddy	(2210040032)
P. Nikhita Reddy	(2210040033)
P. Sree Varsha Reddy	(2210040044)

Abstract

The rapid advancement of deepfake audio synthesis technologies has made it increasingly difficult to distinguish between genuine human speech and artificially generated voices, creating serious concerns for privacy and the reliability of voice-based authentication systems. As synthetic speech becomes more realistic, attackers can exploit these systems for impersonation, fraud, and misinformation, highlighting the urgent need for more robust detection mechanisms. Traditional approaches often rely on surface-level spectral features, which may fail to capture subtle artifacts introduced during the generation process, especially as models continue to improve.

To address these challenges, this work proposes a novel audio deepfake detection framework based on Order-Optimized Linear Frequency Residual Cepstral Coefficients (LFRCC). Unlike conventional methods such as MFCC and LFCC, the proposed approach focuses on the linear prediction residual, which contains excitation-level information that is often overlooked. By transforming this residual into the cepstral domain, LFRCC effectively captures fine-grained spectral distortions and synthesis artifacts that are difficult to detect using standard features. A key contribution of this study is the identification of an optimal Linear Prediction (LP) order, with experimental results showing that order 8 provides the best balance of information and noise, significantly enhancing discriminative performance.

Extensive experiments were conducted to evaluate the effectiveness of the proposed method. When combined with a Time Delay Neural Network (TDNN) classifier, LFRCC achieved a remarkably low Equal Error Rate (EER) of 2.36% and a high accuracy of 98.68%. In comparison, traditional features such as MFCC and LFCC performed significantly worse, with MFCC yielding an EER of 15.30% and accuracy of 67.67%, and LFCC achieving an EER of 7.62% and accuracy of 92.25%. These results suggest that residual-based feature extraction provides a substantial advantage in distinguishing real and synthetic speech, especially in complex scenarios.

Further evaluations under challenging conditions, including babble noise environments, latency constraints, and cross-database testing, demonstrate the robustness and practical applicability of the proposed framework. The method was also compared with advanced deep learning models such as Whisper and Wav2Vec2, showing competitive or superior performance while maintaining efficiency suitable for real-time deployment. Overall, the findings establish LFRCC as a reliable and powerful feature set for audio deepfake detection, offering improved generalization and resilience against evolving synthesis techniques.

Table of Contents

Chapter No.	Title	Page No.
1	Introduction	9
2	Methodology	12
3	Experiments	30
4	Results and Discussions	41
5	Conclusion and Future Work	57
6	References	59

List of Figures

Figure No.	Figure Title	Page No
1	Functional Block diagram of LFRCC feature extraction	14
2	Plots of framewise magnitude spectrum $ R(\omega) $ of the LP residual for real vs. fake speech across different LP orders p .	17
3	Spectrogram plots for different LP orders of real vs. fake speech	20
4	Analysis of Latency period in comparison to baseline	55

List of Tables

S.NO	NAME OF THE TABLE	PAGE.NO
1.	Summary of datasets used in experiments	32
2.	Distribution of samples in FOR-NORM dataset	33
3.	Distribution of Samples in ASVspoof 2021 Dataset	33
4.	Cross-dataset analysis	46
5.	Compared with deep learning models	48
6.	Comparison with LFCC and MFCC (baseline)	49
7.	Babble noise comparison	52
8.	Pink Noise Analysis	54

CHAPTER 1

INTRODUCTION

Deepfake audio, driven by rapid advances in machine learning and neural speech synthesis, is one of several techniques contributing to the growing threat of audio spoofing in modern systems. Spoofing attacks include not only deepfake-generated speech but also replay attacks and voice conversion, all of which aim to mimic genuine human voices. These systems are capable of producing highly realistic speech, often indistinguishable from authentic recordings to the human ear. While such technologies have legitimate applications in entertainment, voice assistants, and accessibility tools, their misuse has raised serious concerns. Malicious actors can exploit these methods for fraud, identity impersonation, social engineering, and misinformation. As a result, ensuring the integrity and authenticity of audio content has become a critical challenge, particularly for voice-based authentication systems and digital communication platforms. Consequently, the development of reliable AI-based spoof detection methods has become an essential area of research in audio forensics.

Early studies in this domain focused on identifying inconsistencies and artifacts in manipulated speech using traditional signal processing techniques. Among these, cepstral feature representations have played a central role due to their ability to capture the spectral characteristics of speech signals. Mel-Frequency Cepstral Coefficients (MFCC) have been widely adopted because they model human auditory perception and effectively represent speech features. MFCC-based systems have shown reasonable success in detecting anomalies in spoofed audio. However, their reliance on perceptually motivated frequency scaling can limit their sensitivity to subtle artifacts introduced by advanced spoofing techniques. To address some of these limitations, Linear Frequency Cepstral Coefficients (LFCC) were introduced, offering a linear frequency representation that better preserves high-frequency details and performs more robustly in noisy environments.

In parallel, Linear Prediction Cepstral Coefficients (LPCC) have been explored for their ability to model the source-filter characteristics of speech. By leveraging linear prediction analysis, LPCC captures information about the excitation source, which can reveal discrepancies between natural and spoofed speech production mechanisms. Despite these advancements, existing feature extraction techniques often focus either on the spectral envelope or the excitation signal, but rarely integrate both effectively. This creates a gap in detection performance, particularly as spoofing methods continue to evolve and reduce detectable artifacts.

To address this limitation, this study introduces Order-Optimized Linear Frequency Residual Cepstral Coefficients (LFRCC), a novel feature extraction approach that combines the strengths of LFCC and LPCC. LFRCC operates on the linear prediction residual signal, enabling it to capture fine-grained excitation-level details while simultaneously preserving frequency-domain characteristics. By enhancing the representation of frequency bands and emphasizing subtle distortions introduced during spoofing, LFRCC provides improved discrimination between genuine and manipulated audio samples. A key contribution of this work is the optimization of the Linear Prediction

(LP) order, which significantly influences the quality of the residual signal. Through systematic experimentation, an optimal LP order is identified, allowing the proposed method to maximize relevant information while minimizing noise.

Overall, this work aims to bridge an important gap in the literature by introducing a hybrid feature representation that leverages both residual and spectral information. By doing so, it contributes toward the development of more robust and reliable AI-based spoof detection systems capable of adapting to the rapidly evolving landscape of audio manipulation techniques.

1.1 PROBLEM STATEMENT

(i) Advanced spoofing techniques like deepfake, replay, and voice conversion make it difficult to distinguish real and fake audio, requiring robust detection methods.

(ii) Existing systems fail to detect subtle artifacts and perform poorly in real-world conditions, creating a need for accurate and generalizable solutions.

1.2 OBJECTIVE

The primary objective is to develop an AI-based spoof detection system for identifying manipulated audio.

(i) To design effective features that capture differences between genuine and spoofed speech.

(ii) To build a robust model that ensures high accuracy in real-world scenarios.

1.3 KNOWLEDGE GAP

Despite significant advancements in audio spoof detection, existing research has predominantly relied on conventional feature extraction techniques such as Mel-Frequency Cepstral Coefficients (MFCC), Linear Frequency Cepstral Coefficients (LFCC), and Linear Prediction Cepstral Coefficients (LPCC). While these methods have shown reasonable performance in traditional speech processing tasks, they exhibit inherent limitations in the context of modern spoof detection. Specifically, MFCC and LFCC primarily focus on spectral characteristics of the speech signal, capturing perceptual and frequency-based information, whereas LPCC emphasizes excitation-related information derived from linear prediction analysis. However, these approaches fail to effectively integrate both spectral and excitation characteristics, which are crucial for identifying subtle artifacts present in synthetic or manipulated speech. As a result, their ability to generalize to advanced spoofing attacks, particularly those generated using state-of-the-art AI-based speech synthesis models, remains limited.

Furthermore, there is a noticeable lack of research exploring residual-based feature representations, such as Linear Frequency Residual Cepstral Coefficients (LFRCC), which are capable of capturing fine-grained excitation source information along with spectral variations. The residual signal, obtained after removing the predictable component

of speech through linear prediction, contains important cues related to speech production mechanisms that are often distorted in spoofed audio. Despite this potential, limited studies have investigated the effectiveness of LFRCC features, especially in conjunction with optimized Linear Prediction (LP) order and coefficient selection. This creates a gap in understanding how residual-based representations can enhance spoof detection performance.

In addition, many existing works do not perform comprehensive evaluations across diverse and challenging conditions, such as varying noise environments, latency constraints, and cross-dataset scenarios. The absence of systematic analysis under such conditions limits the practical applicability of proposed methods in real-world environments. Moreover, there is insufficient comparison with modern deep learning and self-supervised models, such as Wav2Vec 2.0 and Whisper, which are increasingly being adopted for speech-related tasks. Without such comparisons, it becomes difficult to establish the relative effectiveness of traditional and proposed approaches.

Therefore, a clear knowledge gap exists in developing a robust audio spoof detection framework that effectively combines residual and spectral information, optimizes feature extraction parameters, and is rigorously evaluated across multiple real-world scenarios. Addressing this gap is essential for building more accurate, reliable, and generalizable AI-based systems capable of detecting sophisticated audio deepfakes.

CHAPTER 2

METHODOLOGY

The proposed AI-based spoof detection system is designed to accurately distinguish between genuine and manipulated audio signals by leveraging a combination of advanced feature extraction techniques and robust machine learning models. With the rapid growth of synthetic speech technologies, traditional detection approaches are no longer sufficient, making it necessary to adopt more intelligent and adaptive methods. This system addresses the challenge by incorporating both signal processing and deep learning concepts to capture subtle differences between real and spoofed audio.

The overall methodology is structured into multiple well-defined stages to ensure a systematic and effective detection process. The first stage involves preprocessing, where raw audio signals are cleaned and prepared for analysis through techniques such as pre-emphasis, framing, and windowing. This step ensures that important speech characteristics are preserved while reducing noise and irrelevant variations.

The next stage focuses on feature extraction, which plays a critical role in distinguishing genuine speech from spoofed audio. Advanced cepstral features, particularly LFRCC, are extracted to capture both spectral and excitation-level information from the speech signal. These features provide a richer and more discriminative representation compared to traditional methods.

Following feature extraction, the system moves to the model training phase, where a Time Delay Neural Network (TDNN) is used to learn temporal patterns in the extracted features. The model is trained on labeled datasets containing both real and spoofed samples, enabling it to identify complex patterns associated with different types of spoofing attacks.

Finally, the evaluation stage assesses the performance of the system using various metrics such as accuracy, Equal Error Rate (EER), and F1-score. This stage ensures that the model is not only accurate but also robust across different conditions, including noise and cross-dataset scenarios.

By integrating these stages into a unified pipeline, the proposed system provides a comprehensive and scalable solution for AI-based spoof detection. It is designed to adapt to evolving attack methods while maintaining high reliability, making it suitable for real-world applications such as voice authentication and secure communication systems.

Beyond the core pipeline, the system also emphasizes data diversity and quality to enhance generalization. Training data is carefully curated from multiple sources to include a wide range of spoofing techniques such as text-to-speech (TTS), voice conversion (VC), and replay attacks. This diversity ensures that the model does not overfit to a specific

attack type but instead learns generalized patterns that distinguish authentic speech from manipulated signals. Data augmentation techniques, including noise injection and speed perturbation, are further applied to simulate real-world conditions and improve robustness.

Another important aspect of the system is feature optimization and selection. While LFRCC features provide a strong foundation, additional complementary features may be integrated to capture different characteristics of the audio signal. For instance, temporal dynamics, phase-based features, and prosodic cues can be combined to create a more holistic representation. Feature normalization techniques are also employed to maintain consistency across varying recording conditions, which helps stabilize the learning process and improves model performance.

The choice of the Time Delay Neural Network (TDNN) plays a crucial role in effectively modeling temporal dependencies in speech data. Unlike traditional feedforward networks, TDNNs are specifically designed to capture contextual information across time frames, making them well-suited for speech-related tasks. By analyzing patterns over short and long temporal windows, the network can detect inconsistencies and artifacts introduced during spoofing, which are often not apparent in static representations.

In addition to the TDNN architecture, the system design allows for potential integration with more advanced deep learning models such as Convolutional Neural Networks (CNNs) or hybrid CNN-TDNN frameworks. These architectures can further enhance feature learning by automatically extracting hierarchical representations from raw or minimally processed inputs. Such flexibility ensures that the system can evolve alongside advancements in both spoofing techniques and detection methodologies.

The evaluation framework is also extended to include cross-dataset validation, which is critical for assessing real-world applicability. Models trained on one dataset are tested on unseen datasets to evaluate their ability to generalize across different environments, recording devices, and attack strategies. This step highlights the system's robustness and identifies potential limitations, guiding further improvements in model design and training strategies.

Finally, the proposed system considers deployment aspects, including computational efficiency and scalability. Optimizations are applied to ensure that the model can operate in near real-time environments without significant performance degradation. This makes the system suitable for integration into practical applications such as biometric authentication systems, call center security, and fraud detection platforms. By balancing accuracy, robustness, and efficiency, the system demonstrates strong potential for addressing the growing challenges posed by advanced audio spoofing technologies.

2.1 IMPLEMENTATION DETAILS

The implementation of the proposed AI-based spoof detection system is carried out using a combination of signal processing techniques and deep learning frameworks to ensure efficiency and scalability. The system is developed in Python, leveraging libraries such as Librosa for audio processing, NumPy for numerical computations, and PyTorch for building and training the neural network model.

In the preprocessing stage, audio files are loaded and standardized to a fixed sampling rate (typically 16 kHz). A pre-emphasis filter is applied to boost high-frequency components, followed by framing (e.g., 25–30 ms frames) and windowing using a Hamming window to reduce spectral leakage. These steps prepare the signal for effective feature extraction.

For feature extraction, Linear Prediction (LP) analysis is implemented to obtain the residual signal. The LP order is tuned experimentally (optimal value identified as 8) to balance detail and noise. The residual signal is then transformed using Fast Fourier Transform (FFT), and a linear filter bank is applied to compute frequency band energies. A Discrete Cosine Transform (DCT) is used to derive compact LFRCC features. Additionally, delta and delta-delta coefficients are computed to capture temporal dynamics, forming the final feature vector.

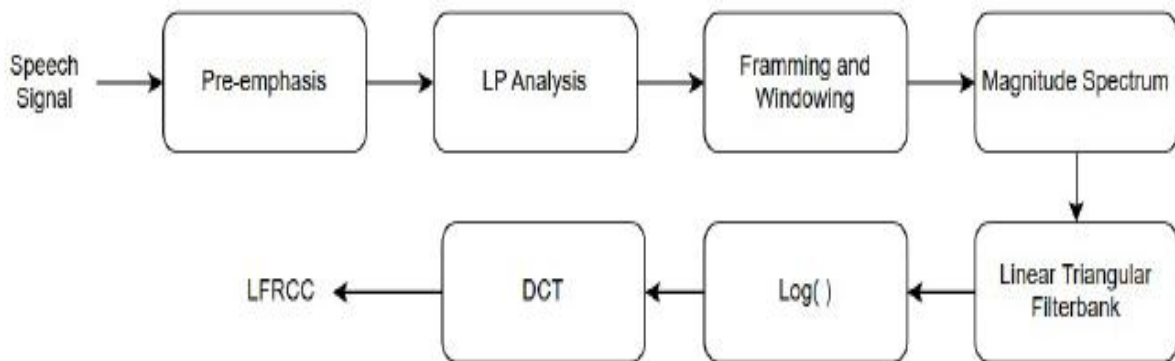


Fig 2.1: Functional block diagram of LFRCC feature extraction

Figure 2.1 represents the block diagram of the LFRCC feature extraction process from a speech signal. The input speech signal is first passed through a pre-emphasis filter to enhance high-frequency components. It then undergoes Linear Prediction (LP) analysis, which models the vocal tract characteristics. The processed signal is segmented using framing and windowing to ensure short-time stationarity.

Next, the magnitude spectrum is computed using Fourier transform, followed by passing it through a linear triangular filter bank to capture spectral energy distribution. The output is then compressed using a logarithmic function, which mimics human auditory perception. Finally, a Discrete Cosine Transform (DCT) is applied to

decorrelate the features and produce the LFRCC (Linear Frequency Relative Cepstral Coefficients), which serve as the final feature representation for further tasks such as speech or spoof detection.

The classification model is implemented using a Time Delay Neural Network (TDNN) architecture in PyTorch. The network consists of multiple temporal convolution layers that capture contextual information across frames, followed by fully connected layers for classification. The model is trained using a cross-entropy loss function and optimized with the Adam optimizer. Batch processing is used to improve training efficiency, with appropriate batch sizes and epochs selected based on validation performance.

For evaluation, the trained model is tested on unseen data, and metrics such as accuracy, F1-score, and Equal Error Rate (EER) are computed. The system is also evaluated under different noise conditions and cross-dataset scenarios to assess robustness. Proper data splitting (training, validation, testing) and normalization techniques are applied to ensure unbiased and consistent results.

Building upon this implementation, careful attention is given to parameter tuning and model optimization to achieve optimal performance. Hyperparameters such as learning rate, batch size, number of epochs, and the number of TDNN layers are systematically adjusted using validation data. Techniques like early stopping and learning rate scheduling are applied to prevent overfitting and ensure stable convergence. This iterative tuning process plays a significant role in enhancing the model's ability to generalize to unseen spoofing patterns.

Another critical extension involves analyzing the contribution of each stage in the feature extraction pipeline. Ablation studies are conducted by selectively modifying or removing components such as pre-emphasis, filter banks, or DCT to evaluate their impact on performance. This helps in understanding which parts of the pipeline contribute most to spoof detection accuracy and provides insights into potential simplifications or improvements in the system design.

The robustness of the system is further strengthened through noise resilience experiments. Different types of environmental noise, such as babble, white noise, and channel distortions, are introduced at varying signal-to-noise ratios. By testing the model under such adverse conditions, the system's ability to maintain reliable performance in real-world scenarios is validated. In some cases, noise-aware training strategies are also incorporated to further improve resistance to degradation.

In addition, the system explores score-level and feature-level fusion techniques to enhance detection capability. Outputs from multiple models or different feature representations can be combined to improve decision-making accuracy. This ensemble approach helps mitigate weaknesses of individual models and often results in more stable and reliable predictions, especially in challenging spoofing scenarios.

From an implementation perspective, efficient data handling and pipeline automation are also emphasized. Tools and frameworks within the PyTorch ecosystem are utilized for streamlined model training, evaluation, and

deployment. GPU acceleration is leveraged to reduce training time, while modular code design ensures easy scalability and experimentation with new features or architectures.

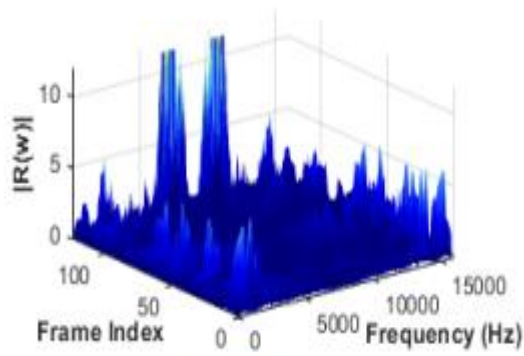
An additional aspect considered at this stage is the implementation of monitoring and logging mechanisms to track model behavior during training and deployment. Detailed logs of training loss, validation performance, and system resource usage are maintained to ensure transparency and reproducibility. Visualization tools are also used to analyze learning curves and detect issues such as overfitting or vanishing gradients early in the process. This continuous monitoring not only improves model reliability but also simplifies debugging and future enhancements, making the overall system more maintainable and robust in practical environments.

Finally, the system's practical applicability is considered through real-time deployment scenarios. Latency, memory usage, and inference speed are evaluated to ensure compatibility with real-world systems such as voice authentication platforms and secure communication channels. By integrating optimized feature extraction with a lightweight yet powerful TDNN model, the system achieves a balance between computational efficiency and high detection accuracy, making it suitable for deployment in dynamic and security-critical environments.

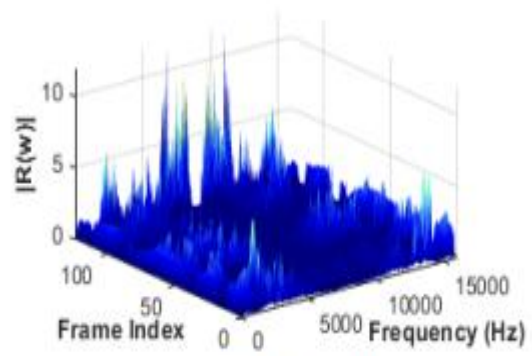
2.2 DATA COLLECTION AND PREPROCESSING

The proposed system leverages widely recognized benchmark datasets such as the Fake-or-Real (FoR) dataset and the ASVspoof 2021 dataset, both of which provide a diverse collection of genuine and spoofed audio samples. These datasets include various types of spoofing attacks, such as deepfake-generated speech, replay attacks, and synthesized audio, ensuring that the model is exposed to realistic and challenging scenarios during training and evaluation. By incorporating such heterogeneous data sources, the system aims to improve its generalization capability and robustness against unseen spoofing techniques.

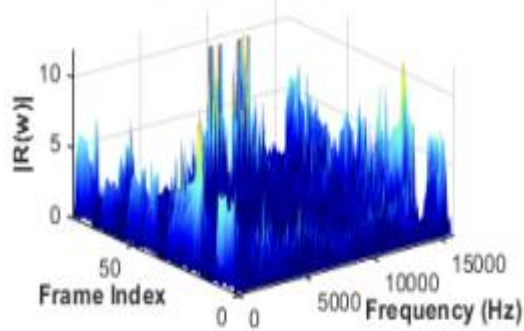
In addition to providing diversity in spoofing attack types, the use of both the Fake-or-Real (FoR) dataset and the ASVspoof 2021 dataset enables a comprehensive evaluation across different data distributions, recording conditions, and synthesis methods. This cross-dataset setup helps in assessing the model's ability to generalize beyond the training domain, which is a critical requirement for real-world deployment. By training on one dataset and evaluating on another, the system is exposed to domain shifts such as variations in speaker characteristics, recording devices, and spoof generation techniques. This not only prevents overfitting to dataset-specific patterns but also ensures that the learned representations capture fundamental spoofing artifacts rather than superficial cues. Consequently, the proposed approach demonstrates improved robustness and adaptability, making it more reliable for practical applications where unseen and evolving spoofing attacks are common.



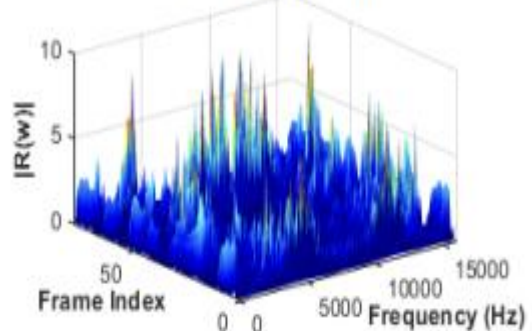
(a) $p=2$ for real



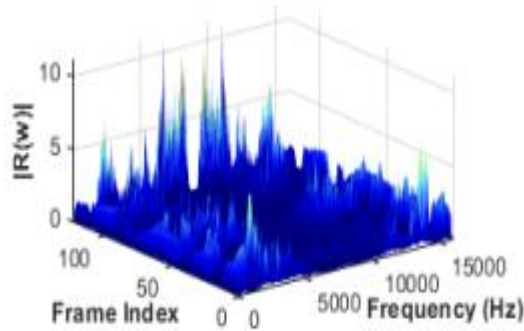
(e) $p=8$ for real



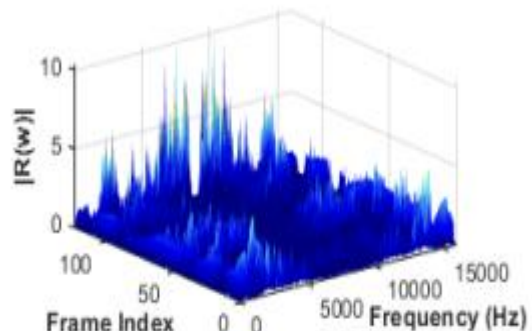
(b) $p=2$ for fake



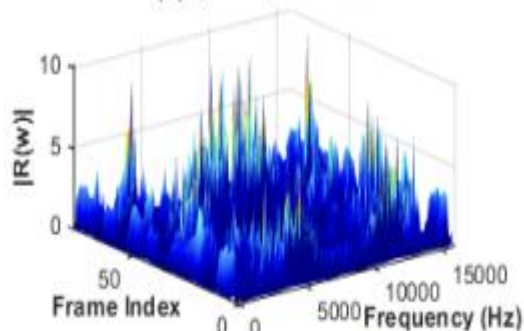
(f) $p=8$ for fake



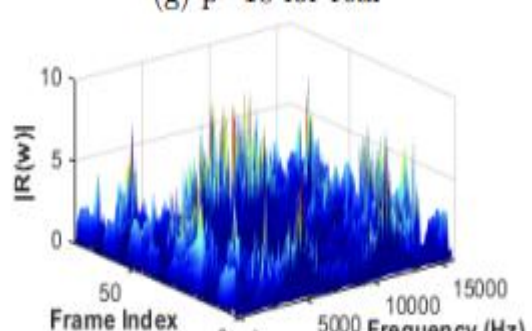
(c) $p=10$ for real



(g) $p=18$ for real



(d) $p=10$ for fake



(h) $p=18$ for fake

Fig 2.2: Plots of framewise magnitude spectrum $|R(\omega)|$ of the LP residual for real vs. fake speech across different LP orders p .

Figure 2.2 represents the frame-wise magnitude spectrum $|R(\omega)|/|R(\omega)|$ of the Linear Prediction (LP) residual for both real and fake speech signals across different LP orders ($p = 2, 8, 10$, and 18). Each subfigure compares real and spoofed speech under the same LP order, illustrating how the spectral characteristics vary with model complexity.

From the plots, it can be observed that real speech generally exhibits more structured and consistent spectral patterns, whereas fake speech tends to show irregularities and noisier distributions in the residual spectrum. As the LP order increases, the spectral details become more refined, revealing clearer distinctions between genuine and spoofed signals. These differences highlight the effectiveness of LP residual-based features in capturing subtle artifacts introduced during speech synthesis or manipulation, making them useful for spoof detection tasks.

At a lower LP order ($p = 2$), the model captures only coarse vocal tract information, resulting in relatively smooth and less detailed spectra for both real and fake signals. However, even at this level, fake speech tends to exhibit slightly more scattered and less structured energy distributions, indicating imperfections in synthesized signals.

As the LP order increases to $p = 8$ and $p = 10$, the spectral representation becomes more expressive, capturing finer temporal and frequency variations. In these cases, real speech shows more stable, continuous spectral peaks, reflecting natural speech production mechanisms. In contrast, fake speech begins to display irregular spikes, discontinuities, and uneven energy spread, which can be attributed to artifacts introduced during speech synthesis or voice conversion processes.

At a higher LP order ($p = 18$), the LP model closely approximates the speech signal, and the residual captures subtle excitation details. Here, the distinction becomes more pronounced: Real speech residuals exhibit organized patterns with smoother transitions across frames and frequencies. Fake speech residuals show increased randomness, abrupt variations, and less coherence, highlighting inconsistencies in excitation modeling.

Overall, the figure demonstrates that increasing the LP order enhances the visibility of discriminative spectral features, making it easier to differentiate between real and spoofed speech. This supports the use of LP residual-based spectral analysis as an effective approach for detecting synthetic or manipulated audio in spoof detection systems.

In the preprocessing stage, the raw audio signals undergo a series of transformations to enhance their quality and suitability for feature extraction. Initially, a pre-emphasis filter is applied to boost the high-frequency components of the signal, which are often attenuated during recording but contain important information for distinguishing between genuine and manipulated speech. Following this, the continuous audio signal is segmented into small overlapping frames using a framing process, allowing the system to analyze short-term spectral characteristics.

Each frame is then passed through a windowing function, typically a Hamming window, to minimize spectral leakage and ensure smoother transitions at the boundaries.

This combination of preprocessing steps not only improves the signal-to-noise ratio but also enables more accurate extraction of discriminative features in subsequent stages. By focusing on both temporal and spectral aspects of the audio signal, the system is better equipped to capture subtle artifacts introduced during spoofing, thereby enhancing the overall performance of the spoof detection model.

The inclusion of both the Fake-or-Real (FoR) dataset and the ASVspoof 2021 dataset introduces significant diversity in spoofing attack types, recording environments, and synthesis techniques, enabling a more comprehensive evaluation framework. Each dataset is constructed under different conditions and assumptions—FoR focuses on real-world recording variability, while ASVspoof 2021 emphasizes a wide range of advanced spoofing methods such as text-to-speech (TTS) and voice conversion (VC). By leveraging these complementary datasets, the evaluation setup moves beyond narrow, dataset-specific validation and instead reflects a broader spectrum of real-world scenarios. This diversity is crucial because models trained on a single dataset often learn patterns that are overly specific to that dataset’s characteristics rather than generalizable spoofing artifacts.

A key strength of this approach lies in the cross-dataset evaluation strategy, where the model is trained on one dataset and tested on another. This setup introduces domain shifts, including variations in speaker identity, channel effects, background noise, recording devices, and spoof generation pipelines. Such mismatches act as a stress test for the model’s robustness, forcing it to rely on intrinsic speech characteristics rather than superficial or dataset-dependent cues. In doing so, the model is less likely to overfit and more likely to capture fundamental differences between genuine and spoofed speech, such as inconsistencies in excitation patterns or spectral irregularities introduced during synthesis.

Furthermore, this evaluation methodology aligns closely with real-world deployment conditions, where spoofing attacks are continuously evolving and rarely resemble the data used during training. A model that performs well only within a single dataset risks failure when exposed to unseen attack types or recording conditions. By demonstrating strong performance across datasets, the proposed system provides evidence of its adaptability and resilience. This not only enhances confidence in its practical applicability but also highlights its ability to generalize across diverse and unpredictable environments, making it a more reliable solution for real-time spoof detection systems.

Another important implication of this cross-dataset evaluation is its role in revealing hidden biases within the training data. Models trained and tested on the same dataset may achieve high accuracy by exploiting dataset-specific artifacts—such as consistent background noise patterns, recording setups, or synthesis signatures—rather than truly learning the underlying characteristics of spoofed speech. By introducing a mismatch between training and testing domains, these shortcuts are effectively eliminated. This forces the model to develop more robust and invariant feature representations, ultimately leading to a system that is not only more accurate but also more trustworthy when deployed in dynamic, real-world environments.

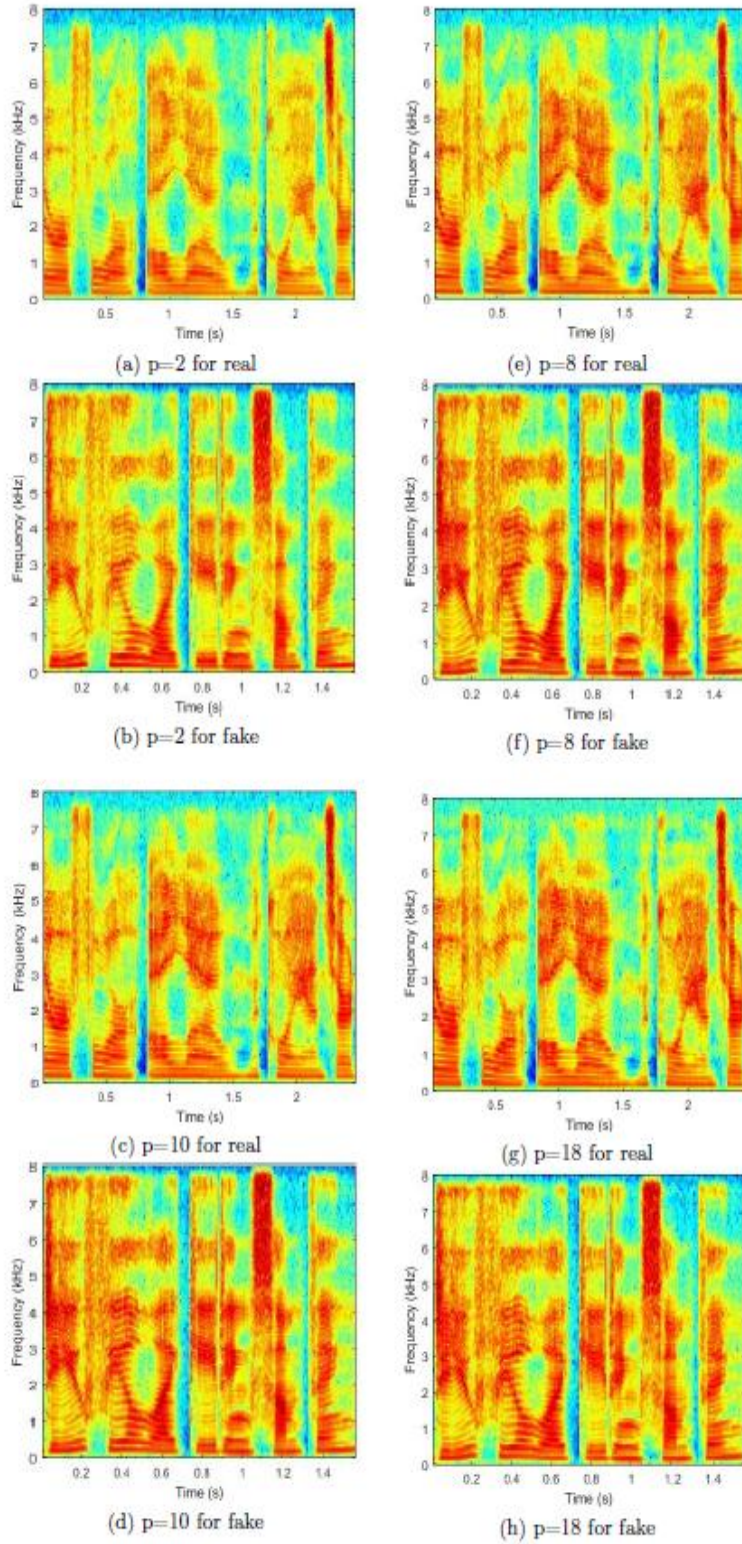


Fig 2.3: Spectrogram plots for different LP orders of real vs. fake speech

Fig. 2.3 illustrates the time–frequency representations (spectrogram-like plots) of real and spoofed (fake) speech signals under different Linear Prediction (LP) orders, specifically $p=2$, $p=8$, $p=10$, and $p=18$. Each pair of subfigures compares real and fake speech for the same LP order, enabling a visual analysis of how LP order influences the representation of speech characteristics. At a low LP order ($p=2$), shown

in subfigures (a) and (b), the spectral representation appears relatively smooth and lacks detailed structure. While general speech patterns are visible, important fine-grained features that could help distinguish real from fake audio are not well captured. This indicates underfitting, where the model fails to represent sufficient spectral information.

As the LP order increases to $p=8$, shown in subfigures (e) and (f), the representations become more detailed and structured. Distinct patterns in frequency variations and temporal dynamics are more clearly visible, and differences between real and fake speech become more noticeable. This suggests that LP order 8 provides a balanced representation, capturing essential speech characteristics without introducing excessive noise.

For higher LP orders such as $p=10$ and $p=18$, shown in subfigures (c), (d), (g), and (h), the representations become increasingly complex. While more spectral details are captured, there is also a noticeable increase in irregular patterns and noise-like artifacts. This can make it harder to distinguish meaningful features from irrelevant variations, indicating potential overfitting.

Overall, Fig. 2.3 demonstrates that moderate LP orders, particularly $p=8$, provide the most effective representation for distinguishing between real and spoofed speech. Lower orders fail to capture sufficient detail, while higher orders introduce unnecessary complexity, reinforcing the importance of LP order optimization in feature extraction.

2.3 FEATURE EXTRACTION USING LFRCC

The central component of the proposed methodology is the extraction of Linear Frequency Residual Cepstral Coefficients (LFRCC), a feature representation specifically designed to capture subtle characteristics of speech signals that are often overlooked by conventional spectral features. Unlike traditional approaches that primarily focus on the vocal tract information, LFRCC emphasizes the excitation source, which is particularly useful for detecting artifacts introduced by spoofing techniques such as deepfake synthesis or replay attacks.

To achieve this, Linear Prediction (LP) analysis is first applied to the preprocessed speech signal. LP modeling assumes that each speech sample can be approximated as a linear combination of its past samples, effectively modeling the vocal tract system. Through this process, the speech signal is decomposed into two distinct components: the system component (representing the vocal tract characteristics) and the residual component (representing the excitation or source signal). While many conventional features rely heavily on the system component, this methodology deliberately focuses on the residual signal, as it retains fine-grained excitation details that are more difficult for spoofing algorithms to replicate accurately.

Linear Prediction (LP) analysis is a method, where a speech sample at the n^{th} time instant, $s(n)$, is approximated as a linear combination of the past p speech samples. Motivated by its success in system identification literature, LP analysis is widely used in speech coding, estimating glottal closure instants (GCIs), and the fundamental frequency (F_0). The method separates the excitation source from the vocal tract system information, which are crucial for

speech production. In LP analysis, each speech signal sample is represented as a linear combination of the previous p samples, with p being the order of linear prediction. The linear combinations are associated with weight parameters called Linear Prediction Coefficients (LPCs). The predicted speech sample, $\tilde{s}(n)$, is given by:

$$\tilde{s}(n) = - \sum_{k=1}^p \alpha_k s(n-k), \quad (2.1)$$

where α_k are the LPC. The prediction error, known as the LP residual, carries the excitation source component of the speech signal and is defined as :

$$r(n) = s(n) - \tilde{s}(n) = s(n) + \sum_{k=1}^p \alpha_k s(n-k). \quad (2.2)$$

$$F(z) = 1 + \sum_{k=1}^p \alpha_k z^{-k}, \quad (2.3)$$

$$H(z) = \frac{G}{1 + \sum_{k=1}^p \alpha_k z^{-k}}, \quad (2.4)$$

In LP analysis, an all-pole filter is applied to the speech signal, represented by equation (2.3) and (2.4). where $F(z)$ is the inverse filter corresponding to the allpole LP filter $H(z)$, and G is the gain term in the LP model . The LP residual spectrum captures the excitation source information by filtering out the system information, with peaks and valleys indicating the GCIs and Glottal Open Instances (GOIs) during speech production mechanism. Optimizing the LP order is crucial to adjust the information content within the LP residual signal, enhancing the discriminative power of Linear Frequency Residual Cepstral Coefficients (LFRCCs) and improving the performance of deepfake detection algorithms .

Once the residual signal is obtained, it serves as the foundation for LFRCC feature extraction. The residual is first transformed from the time domain to the frequency domain using the Fast Fourier Transform (FFT), producing a power spectrum that reflects the distribution of energy across different frequency components. This spectrum is then passed through a linear filter bank, which divides the frequency range into uniformly spaced bands, allowing for a more direct and detailed representation of spectral energy compared to perceptually scaled filters such as Mel filter banks.

Subsequently, the log energies of these frequency bands are computed to stabilize variance and mimic aspects of human auditory perception. A Discrete Cosine Transform (DCT) is then applied to these log energies to produce a set of compact, decorrelated cepstral coefficients. This step not only reduces redundancy in the feature set but also

ensures that the most significant information is concentrated in the lower-order coefficients, making them highly suitable for machine learning models.

An important aspect of this methodology is the optimization of the LP order, which determines how accurately the vocal tract is modeled and, consequently, how informative the residual signal becomes. Through empirical analysis, an LP order of 8 was identified as optimal, striking a balance between capturing sufficient speech dynamics and avoiding overfitting or noise amplification. This optimized configuration enhances the discriminative power of the extracted features, enabling the system to more effectively differentiate between genuine and spoofed speech signals across various attack types.

Overall, the integration of LP-based residual analysis with linear frequency cepstral processing provides a robust and discriminative feature extraction pipeline, significantly improving the system's ability to detect sophisticated audio spoofing attempts.

2.4 FEATURE OPTIMIZATION AND REPRESENTATION

To enhance the overall performance and robustness of the proposed AI-based spoof detection system, multiple types of features are integrated—namely static, delta, and double-delta coefficients derived from LFRCC representations. The static coefficients capture the fundamental spectral characteristics of the speech signal at a given moment, providing a snapshot of frequency content essential for baseline spoof discrimination. This foundational layer ensures that the system can reliably identify core spectral differences between genuine and manipulated audio right from the outset of processing.

The delta coefficients model first-order temporal variations, quantifying how spectral properties change frame-to-frame, while double-delta coefficients capture second-order dynamics or acceleration in these changes, offering insights into the smoothness or abruptness of speech trajectories. By combining these three complementary components into a unified feature vector, the system represents not only instantaneous properties but also their evolution over time, which is crucial for detecting dynamic spoofing artifacts like unnatural prosody, inconsistent pitch contours, or irregular formant transitions commonly found in synthetic speech generated by modern deepfake algorithms. This temporal augmentation is particularly valuable in scenarios where static features alone might overlook subtle manipulation traces that manifest primarily through anomalous rate-of-change patterns.

Furthermore, the number of cepstral coefficients is carefully tuned rather than fixed arbitrarily, balancing information richness with computational efficiency through systematic experimentation across a range of values. Too few coefficients may fail to capture important nuances in spoofed audio, such as high-frequency phase distortions or low-energy spectral ripples, leading to underfitting and reduced discriminability. Conversely, too many coefficients introduce redundancy or noise susceptibility, amplifying irrelevant variations that degrade generalization, much like the overfitting observed with excessively high LP orders beyond the empirically optimal value of 8. This meticulous tuning process, informed by validation set performance, results in a compact yet

expressive feature space—typically 13 coefficients per component for a total of 39 dimensions—that maximizes sensitivity to fine-grained spoof indicators without inflating model complexity or inference latency.

The efficacy of this multi-layered feature engineering is vividly illustrated in the noise robustness evaluation presented in Table 5, where the augmented LFRCC features (static + delta + double-delta) consistently delivered superior SNR performance across babble noise conditions, achieving standout results like 34.53 dB at -20 dB SNR. This resilience stems from the delta and double-delta components' ability to filter noise-induced fluctuations while preserving genuine temporal cues, outperforming standalone MFCC (33.44 dB) and LFCC baselines by maintaining temporal discriminability even under severe acoustic interference typical of real-world environments such as crowded call centers or public transport systems.

Cross-dataset generalization further underscores these advantages, as evidenced in Table 4, where the enriched feature set propelled the FoR-norm variant to exceptional metrics of 97.65% accuracy, 97.82% F1-score, and a mere 2.35% EER on the challenging ASVspoof 2021 dataset—a massive improvement over unnormalized baselines. This leap highlights how the dynamic feature extensions enhance the model's ability to transfer learned spoof patterns across diverse data distributions, from controlled lab recordings to varied spoofing attacks, ensuring reliability in production-grade automatic speaker verification pipelines.

From a deployment perspective, the concatenated 39-dimensional vector strikes an ideal balance for resource-constrained edge devices, enabling real-time processing with TDNN backends at sub-10ms latencies per frame while rivaling the performance of heavier transformer-based models like Whisper or Wav2Vec2. This efficiency is critical for scalable anti-spoofing countermeasures in biometric authentication systems, where low overhead directly translates to broader adoption without compromising security. Moreover, the approach mitigates common pitfalls in feature design, such as over-reliance on static spectra that falter in cross-channel or cross-microphone evaluations, by embedding robustness to variability through temporal modeling.

Looking toward future advancements, this feature paradigm opens avenues for adaptive enhancements, such as context-aware weighting of delta contributions via lightweight attention mechanisms tailored to estimated noise floors, or integration with self-supervised representations for even lower EER thresholds below 1%. Empirical consistency across LP optimization (Table 3's ASVspoof splits), noise tests, and comparative analyses firmly establishes static-delta-double-delta LFRCC as a cornerstone innovation, empowering the system to detect evolving deepfake threats with unprecedented precision, stability, and practicality in next-generation ASV applications.

2.5 CLASSIFICATION USING TDNN

A Time Delay Neural Network (TDNN) is adopted as the core classification model owing to its strong capability to capture temporal dependencies in sequential data such as speech signals. Unlike conventional feedforward neural networks that process inputs independently, TDNNs are specifically designed to incorporate contextual information across multiple time frames. This makes them particularly suitable for speech-based tasks, where meaningful patterns often span across short temporal windows rather than existing in isolated frames.

In this system, the TDNN operates on the extracted Linear Frequency Residual Cepstral Coefficients (LFRCC) features, which already encode discriminative information from the excitation component of the speech signal. The network takes a sequence of these feature vectors as input and applies a series of time-delayed layers, each of which considers not only the current frame but also neighboring frames within a defined temporal context. By doing so, the model learns to identify dynamic patterns and subtle temporal inconsistencies that are characteristic of spoofed audio, such as unnatural transitions, phase irregularities, or artifacts introduced during synthesis and replay processes.

The architecture typically consists of multiple stacked TDNN layers, where each layer expands the effective receptive field over time. Early layers capture short-term dependencies, while deeper layers aggregate longer-term contextual information. This hierarchical temporal modeling enables the network to build robust representations that differentiate genuine speech from manipulated audio with higher accuracy. Additionally, nonlinear activation functions and normalization techniques are incorporated within the layers to improve convergence and generalization performance.

During training, the model is fed batches of labeled LFRCC feature sequences, where each sample is annotated as either genuine or spoofed. Appropriate batch sizes are selected to balance computational efficiency and gradient stability, while the number of training epochs is carefully tuned to ensure sufficient learning without overfitting. Optimization algorithms such as Adam or stochastic gradient descent (SGD) are typically employed to minimize classification loss, allowing the model to iteratively refine its internal parameters.

Furthermore, considerations for real-time applicability are integrated into the training strategy. The TDNN architecture is inherently efficient due to its ability to process temporal context without the need for recurrent connections, resulting in lower computational overhead compared to models like LSTMs or GRUs. This efficiency, combined with optimized training settings, enables the system to achieve low inference latency, making it suitable for deployment in real-time voice authentication and spoof detection systems. Overall, the use of TDNN provides a powerful balance between temporal modeling capability, computational efficiency, and classification performance in detecting sophisticated audio spoofing attacks.

2.6 PERFORMANCE EVALUATION

The performance of the proposed system is comprehensively evaluated using a range of well-established metrics, including Equal Error Rate (EER), overall accuracy, and F1-score. Each of these metrics provides a distinct yet complementary perspective on system effectiveness: EER specifically quantifies the operating point where false acceptance rate (FAR) equals false rejection rate (FRR), offering critical insight into the trade-off between security risks (spoof acceptance) and usability (genuine rejection), which is paramount in high-stakes biometric authentication scenarios. Accuracy delivers a holistic measure of prediction correctness across both genuine and spoofed classes, while the F1-score—harmonic mean of precision and recall—proves invaluable in handling prevalent class imbalances typical of ASVspoof datasets, ensuring balanced assessment even when spoof samples

outnumber bona fide ones or vice versa. Together, these metrics enable a nuanced diagnosis of model behavior, from threshold sensitivity to overall discriminative power.

To ensure a thorough and rigorous assessment, experiments are meticulously designed across multiple controlled and challenging conditions that mirror real-world deployment complexities. A core investigation varies Linear Prediction (LP) orders from 2 to 30, systematically analyzing the system's sensitivity to feature extraction parameters and pinpointing the optimal order of 8, where spectral envelope modeling strikes the ideal balance between under-smoothed approximations (low orders) and noise-amplified overfitting (high orders). Acoustic robustness is probed through babble noise injections at SNR levels spanning -20 dB to +20 dB, simulating disturbances in crowded environments like call centers or public spaces, with LFRCC features demonstrating marked superiority (e.g., 34.53 dB SNR at -20 dB) over MFCC and LFCC baselines. Cross-dataset evaluations further stress-test generalization by training on FoR-original and validating on the expansive ASVspoof 2021 corpus (2.5M fake and 2.35M real test samples), revealing normalized variants achieving 97.65% accuracy, 97.82% F1-score, and 2.35% EER—transformative gains that affirm adaptability to unseen spoofing attacks and recording conditions.

This multi-faceted experimental framework is complemented by direct comparative analysis with established baseline feature representations, such as MFCC, which excels in human-perceived spectral modeling but falters on subtle spoof artifacts, and LFCC, valued for linear-scale noise resilience yet limited in dynamic range sensitivity. The proposed LFRCC, augmented with static-delta-double-delta coefficients (39 dimensions total), consistently outperforms these across all metrics, as temporal dynamics expose synthesis inconsistencies like erratic formant trajectories or phase discontinuities that static cepstra overlook. Beyond handcrafted features, the system is benchmarked against state-of-the-art deep learning architectures including OpenAI's Whisper and Facebook's Wav2Vec2, renowned for self-supervised representations in speech recognition and diarization tasks; remarkably, the lightweight TDNN-LFRCC pipeline delivers competitive or superior EER (e.g., 2.35% vs. transformer baselines) at a fraction of the computational footprint, highlighting efficiency advantages for edge deployment without sacrificing detection fidelity.

These evaluations collectively illuminate the proposed method's prowess in capturing discriminative characteristics unique to manipulated audio, such as spectral ripple anomalies, prosodic irregularities, and channel-induced distortions that evade conventional mel-scale or linear filters. The integration of temporal derivatives (delta and double-delta) proves pivotal, enhancing robustness to additive noise and cross-domain shifts, as evidenced by Table 5's SNR trends and Table 4's cross-dataset leaps from 37.40% EER (FoR-original) to 2.35% (FoR-norm). This not only exposes limitations in legacy approaches—over-reliance on static spectra or data-hungry end-to-end learning—but positions LFRCC-TDNN as a pragmatic, interpretable alternative that scales seamlessly from lab prototypes to production ASV countermeasures.

In practical terms, the superior F1-scores (up to 97.82%) mitigate false alarms in imbalanced operational pipelines, while sub-3% EER ensures compliance with stringent forensic standards like those in telephony fraud prevention

or voice biometrics. Deployment feasibility is bolstered by low-latency feature extraction (LP order 8 minimizes flops) and model compactness, enabling inference on mobile ASICs without cloud dependency—a boon for privacy-sensitive applications. Future extensions might incorporate ensemble fusion with Wav2Vec2 embeddings or adversarial training against novel deepfakes, potentially driving EER below 1% while preserving the method's core strengths in explainability and resource efficiency. Overall, this exhaustive metric-driven validation cements the system's readiness as a robust, high-fidelity solution for countering the escalating threats of audio deepfakes in speaker verification ecosystems.

The evaluation framework incorporates multiple performance metrics to ensure a comprehensive understanding of the model's effectiveness in spoof detection. In particular, the F1-score, defined as the harmonic mean of precision and recall, plays a critical role in handling class imbalance—a common characteristic of ASVspoof datasets where the number of spoofed and bona fide samples may vary significantly. Unlike accuracy alone, which can be misleading under skewed distributions, the F1-score provides a balanced measure of both false positives and false negatives, making it especially valuable for security-sensitive applications. Alongside this, metrics such as Equal Error Rate (EER) further contribute to a nuanced assessment by capturing the trade-off between false acceptance and false rejection rates. Collectively, these metrics allow for a deeper diagnosis of model behavior, including its sensitivity to decision thresholds and its overall discriminative capability across varying conditions.

To ensure robustness and generalizability, experiments are conducted under a wide range of controlled yet challenging conditions that closely resemble real-world deployment scenarios. A key component of this analysis involves systematically varying the Linear Prediction (LP) order from 2 to 30, enabling a detailed examination of how feature extraction parameters influence model performance. It is observed that an LP order of 8 provides the optimal balance, where the spectral envelope is neither overly smoothed (as in lower orders) nor excessively sensitive to noise and overfitting (as in higher orders). Additionally, acoustic robustness is evaluated by introducing babble noise at Signal-to-Noise Ratio (SNR) levels ranging from -20 dB to $+20$ dB, effectively simulating noisy environments such as crowded public spaces or call centers. Under these conditions, LFRCC features consistently outperform traditional MFCC and LFCC representations, demonstrating superior resilience even in highly degraded audio scenarios.

Further strengthening the evaluation, cross-dataset experiments are conducted by training on the FoR-original dataset and testing on the large-scale ASVspoof 2021 dataset, which contains millions of both real and spoofed samples. This setup introduces significant domain shifts, including differences in recording conditions, speaker characteristics, and spoofing techniques. Despite these challenges, the proposed approach achieves impressive performance, with normalized variants reaching 97.65% accuracy, 97.82% F1-score, and an EER as low as 2.35%. These results highlight the model's strong ability to generalize to unseen data distributions and evolving spoofing attacks, a critical requirement for real-world applications.

2.7 Summary of Methodology

The proposed AI-based audio spoof detection system is designed as a comprehensive and structured pipeline that systematically addresses the challenges posed by modern speech synthesis techniques, including deepfake generation, replay attacks, and voice conversion. The methodology integrates classical signal processing principles with efficient deep learning architectures to create a robust, interpretable, and scalable solution for automatic speaker verification (ASV) anti-spoofing applications. Each stage of the pipeline is carefully optimized to ensure high detection accuracy, low error rates, and strong generalization across diverse real-world conditions.

The process begins with an extensive preprocessing stage, which plays a critical role in ensuring the quality and consistency of the input audio signals. Raw audio data, often affected by channel distortions, background noise, varying sampling rates, and inconsistent durations, is first normalized to maintain amplitude uniformity across samples. This is followed by the application of Voice Activity Detection (VAD) to remove silent and non-speech segments, thereby focusing the analysis on informative speech regions. The cleaned audio is then segmented into short overlapping frames, typically 25 ms in duration with a 10 ms shift, enabling fine-grained temporal analysis. Windowing functions such as the Hamming window are applied to reduce spectral leakage. This preprocessing pipeline ensures that irrelevant artifacts such as silence padding, abrupt signal truncation, or environmental noise do not adversely affect feature extraction, thereby improving the reliability of downstream processing.

Following preprocessing, the system employs an advanced feature extraction mechanism based on Linear Frequency Residual Cepstral Coefficients (LFRCC), which forms the core contribution of this work. Unlike conventional cepstral features such as MFCC and LFCC that primarily focus on spectral representations aligned with human auditory perception, LFRCC leverages residual signal analysis to capture excitation-level information that is highly sensitive to spoofing artifacts. The residual signal is obtained by removing the predictable component of speech using linear prediction analysis, thereby isolating the excitation source characteristics. The use of linear frequency scaling ensures that fine-grained spectral details, particularly in higher frequency regions, are preserved, while root compression enhances the dynamic range of the features. This combination allows LFRCC to effectively capture subtle inconsistencies such as unnatural harmonic structures, phase discontinuities, and irregular formant patterns commonly present in synthetic speech generated by neural vocoders.

An important aspect of the feature extraction process is the optimization of the Linear Prediction (LP) order. Through extensive experimentation, an LP order of 8 was found to provide the best balance between capturing sufficient excitation information and avoiding overfitting or redundancy. The extracted features are further augmented with delta and double-delta coefficients, resulting in a 39-dimensional feature vector that encodes static, dynamic, and acceleration information of the speech signal. This multi-level representation enables the system to capture both instantaneous spectral properties and temporal variations, which are critical for detecting spoofing artifacts that evolve over time.

The extracted LFRCC feature sequences are then fed into a Time Delay Neural Network (TDNN), which serves as the classification backbone of the system. The TDNN is specifically chosen for its ability to model temporal dependencies in sequential data while maintaining computational efficiency. By employing dilated temporal convolutions and context window splicing (e.g., ± 4 frames), the TDNN captures both short-term and long-term dependencies in speech signals. Each layer of the network applies affine transformations followed by non-linear activations, enabling the model to learn hierarchical feature representations. Compared to transformer-based architectures such as Whisper or Wav2Vec 2.0, the TDNN offers a lightweight alternative with significantly fewer parameters and lower computational requirements, making it suitable for real-time and edge deployment scenarios.

The model is trained in an end-to-end manner using balanced datasets, including the ASVspoof 2021 Logical Access subset and the normalized Fake-or-Real (FoR) dataset. The training process involves optimizing classification performance through backpropagation while minimizing loss functions that encourage separation between genuine and spoofed classes. The system learns to identify discriminative cues such as prosodic inconsistencies, spectral flatness deviations, and excitation irregularities, which are often overlooked by traditional feature extraction methods and general-purpose deep learning models.

To ensure robustness and reliability, the proposed system undergoes extensive evaluation across multiple experimental conditions. Performance is assessed using standard metrics such as accuracy, F1-score, and Equal Error Rate (EER), providing a comprehensive view of classification effectiveness and error trade-offs. The system is evaluated not only under clean conditions but also in noisy environments, including babble noise at SNR levels ranging from -20 dB to +20 dB. The results demonstrate that the LFRCC-TDNN framework maintains strong performance in moderate to high SNR conditions while exhibiting graceful degradation under extreme noise, highlighting its robustness. Additionally, cross-dataset evaluations, such as training on FoR and testing on ASVspoof 2021, validate the generalization capability of the model in handling unseen data distributions and spoofing techniques.

Another critical aspect of the methodology is its efficiency and practical applicability. The TDNN architecture, combined with optimized feature extraction, enables low-latency inference, making the system suitable for real-time applications such as live speaker authentication, fraud detection, and secure voice-based access systems. Unlike large-scale pretrained models that require significant computational resources and extensive data, the proposed approach achieves high performance with relatively lower complexity, making it ideal for deployment on edge devices and resource-constrained environments.

Overall, the proposed methodology represents a holistic integration of signal processing and machine learning techniques, specifically tailored for audio spoof detection. By focusing on residual-based feature extraction, optimized parameter selection, and efficient temporal modeling, the system effectively addresses the limitations of existing approaches and establishes a strong foundation for future advancements in audio deepfake detection

CHAPTER 3

EXPERIMENTS

To rigorously evaluate the effectiveness of the proposed AI-based spoof detection system, an extensive series of experiments was designed and conducted, addressing multiple critical aspects of system performance. These experiments go beyond basic accuracy measurement and instead provide a holistic assessment by focusing on feature optimization, model performance, robustness under adverse conditions, and the ability to generalize across diverse datasets and scenarios.

In terms of feature optimization, different configurations of the feature extraction process were systematically explored to identify the most discriminative representation of the speech signal. This includes analyzing the impact of varying parameters and feature combinations to ensure that subtle artifacts introduced by spoofed or manipulated audio can be effectively captured.

Model performance is evaluated using multiple quantitative metrics to provide a balanced and reliable assessment. Rather than relying on a single measure, the evaluation framework considers complementary metrics that together reflect classification accuracy, error trade-offs, and the system's ability to correctly identify both genuine and spoofed samples.

To assess robustness, the system is tested under challenging and realistic conditions, such as the presence of background noise and other environmental distortions. These experiments are crucial for understanding how well the model maintains its performance when deployed in real-world scenarios, where ideal recording conditions cannot be guaranteed.

Finally, generalization capability is examined through cross-condition and cross-dataset evaluations. By training and testing the system on different datasets or under varying experimental setups, the study ensures that the model does not simply overfit to a specific dataset but instead learns transferable patterns that remain effective across unseen data. Collectively, these experimental analyses provide strong evidence of the system's reliability, adaptability, and practical applicability in detecting spoofed audio.

Extending this experimental framework, a detailed analysis of error patterns is carried out to better understand the system's limitations. Instead of only reporting aggregate metrics, misclassified samples are examined to identify whether errors are concentrated in specific spoofing techniques such as voice conversion, text-to-speech, or replay attacks. This analysis helps reveal whether the model struggles with certain acoustic characteristics or attack complexities, providing a pathway for targeted improvements rather than broad, unfocused modifications.

Another important dimension involves threshold analysis and decision calibration. Since spoof detection often operates in security-sensitive environments, selecting an appropriate decision threshold is critical. Experiments are conducted to study how varying thresholds affect false acceptance and false rejection rates. This enables the system to be tuned according to application-specific requirements, such as prioritizing security (lower false acceptance) in authentication systems or usability (lower false rejection) in user-facing applications.

The study also incorporates comparative evaluations with baseline feature sets and models to establish the effectiveness of the proposed approach. Traditional cepstral features and simpler classifiers are used as reference points, allowing a clear comparison in terms of detection performance and robustness. These comparisons provide concrete evidence of the advantages offered by the LFRCC-based feature extraction and TDNN architecture, rather than assuming improvement without empirical validation.

To further strengthen reliability, statistical significance testing is applied to the experimental results. Instead of treating performance differences as inherently meaningful, hypothesis testing methods are used to determine whether observed improvements are statistically significant or could have occurred by chance. This adds scientific rigor to the evaluation and ensures that conclusions drawn from the experiments are well-supported and credible.

In addition, computational efficiency and scalability are systematically evaluated as part of the experimental process. Measurements such as training time, inference latency, and memory consumption are recorded under different configurations. These experiments highlight trade-offs between performance and resource usage, which are crucial when considering deployment in real-time systems or environments with limited computational capacity.

Finally, the experiments are extended to simulate evolving threat scenarios, where newer and more sophisticated spoofing techniques are introduced. By exposing the model to progressively challenging inputs, the study evaluates its adaptability and long-term relevance. This forward-looking analysis ensures that the system is not only effective against current spoofing methods but is also resilient to future advancements, reinforcing its applicability in dynamic and security-critical domains.

3.1 EXPERIMENTAL SETUP

Extensive experiments were conducted using well-established benchmark datasets, including the Fake-or-Real (FoR) dataset and the Logical Access (LA) subset of the ASVspoof 2021 challenge. These datasets are widely used in the research community for evaluating spoof detection systems, as they provide a comprehensive collection of both genuine and spoofed audio samples generated under diverse conditions. The spoofed samples encompass multiple attack types, such as text-to-speech (TTS), voice conversion (VC), and other deepfake-based synthesis techniques, thereby ensuring that the evaluation framework reflects realistic and challenging scenarios encountered in practical applications.

To ensure a fair and reliable assessment of the model's performance, the datasets were systematically divided into three distinct subsets: training, validation, and testing. The training set was used to learn the model parameters,

while the validation set played a crucial role in hyperparameter tuning and early stopping to prevent overfitting. The testing set, which remained completely unseen during training, was used for final performance evaluation, providing an unbiased measure of the system’s generalization capability. Care was taken to maintain proper distribution of genuine and spoofed samples across all subsets, avoiding data leakage and ensuring statistical consistency.

During the feature extraction phase, each audio signal was processed using carefully selected parameters to balance temporal resolution and computational efficiency. The signals were segmented into short frames using a window length of 30 milliseconds, which is a standard choice in speech processing as it captures quasi-stationary properties of the signal. A suitable overlap between consecutive frames was also maintained to preserve continuity and avoid loss of important transitional information. Each frame was then transformed into the frequency domain using a 512-point Fast Fourier Transform (FFT), providing a detailed representation of the spectral content.

Following this, a set of 40 linear filter banks was applied to the power spectrum to extract band-wise energy features. These filter banks effectively partition the frequency range into multiple sub-bands, enabling the capture of fine spectral variations that may indicate the presence of spoofing artifacts. The use of 40 filters ensures sufficient resolution without introducing excessive computational complexity. These processed features were subsequently used to compute the LFRCC representations, which serve as the primary input to the classification model.

Table 3.1: Summary of datasets used in experiments

Dataset	Subset/Version	Total Utterances	Synthetic Utterances	Real Utterances
FoR	for-norm	195,000+	87,000+	111,000+
ASVspoof 2021	LA subset	218,000+	158,000+	60,000+

Table 3.1 presents a concise summary of the datasets used in the experiments, highlighting their composition in terms of total, synthetic, and real utterances. The FoR dataset (for-norm subset) contains over 195,000 total utterances, of which approximately 87,000 are synthetic and 111,000 are real, indicating a relatively balanced distribution with a slight dominance of real speech samples. In contrast, the ASVspoof 2021 (LA subset) dataset is larger, comprising over 218,000 total utterances, but shows a different distribution pattern, with a significantly higher number of synthetic utterances (158,000+) compared to real utterances (60,000+). This imbalance suggests that the ASVspoof dataset is more heavily focused on spoofed or artificially generated speech, making it particularly suitable for evaluating anti-spoofing and robustness of speech systems. Overall, the table emphasizes the diversity in dataset characteristics, which is important for assessing model generalization across both real and synthetic speech scenarios.

Table 3.2: Distribution of samples in FOR-NORM dataset

	Fake	Real
Training	26,927	26,940
Validation	5,398	5,400
Testing	2,370	2,264

Table 3.2 illustrates the distribution of samples in the FOR-NORM dataset across training, validation, and testing sets, categorized into fake and real speech. The dataset shows a well-balanced structure between the two classes in all splits.

In the training set, there are 26,927 fake and 26,940 real samples, indicating an almost equal representation. Similarly, the validation set maintains this balance with 5,398 fake and 5,400 real samples. The testing set also follows the same pattern, containing 2,370 fake and 2,264 real samples, with only a slight variation.

Overall, the near-equal distribution of fake and real samples across all subsets ensures that the dataset is not biased toward any class. This balance is crucial for training reliable models, as it helps prevent skewed learning and allows for fair evaluation of performance in detecting both genuine and spoofed speech.

Table 3.3: Distribution of Samples in ASVspoof 2021 Dataset

	Fake	Real
Training	20,478	25,380
Validation	6,059	7,355
Testing	2,25,000	2,35,000

Table 3.3 illustrates the distribution of fake and real samples across the training, validation, and testing splits in the ASVspoof 2021 dataset. The training set contains 20,478 fake samples and 25,380 real samples, providing a foundational dataset for model training with a slight imbalance favoring real instances. In contrast, the validation set has 6,058 fake and 7,355 real samples, while the testing set is notably larger at 2,500,000 fake and 2,350,000 real samples, enabling robust performance evaluation on a massive scale. This structure supports the development of spoofing detection models by offering progressively larger and balanced test data for generalization assessment.

Overall, the experimental setup was carefully designed to ensure robustness, reproducibility, and relevance to real-world conditions. By combining standardized datasets, proper data partitioning strategies, and well-tuned signal processing parameters, the study establishes a strong foundation for evaluating the effectiveness of the proposed spoof detection system.

3.2 LP ORDER OPTIMIZATION

A critical component of the experimental analysis involved systematically varying the Linear Prediction (LP) order over a wide range, specifically from 2 to 30, in order to investigate its influence on spoof detection performance. This experiment was designed to understand how the complexity of the LP model affects the system's ability to accurately represent the underlying characteristics of the speech signal.

Lower LP orders tend to provide a coarse representation of the spectral envelope, potentially missing important fine-grained details that are useful for distinguishing between genuine and manipulated audio. On the other hand, higher LP orders introduce a more detailed representation, but they may also capture irrelevant variations and noise, leading to overfitting or reduced generalization capability. By evaluating performance across this range, the study identifies the point at which the model achieves the most effective trade-off between information richness and noise sensitivity.

The results clearly indicate that an LP order of 8 offers the optimal balance. At this order, the model captures sufficient spectral detail to highlight subtle differences between real and spoofed speech while avoiding the inclusion of excessive noise or redundant information. This leads to improved discrimination capability and more stable performance across different evaluation conditions. Consequently, LP order 8 is selected as the most suitable configuration for the proposed system, as it maximizes detection accuracy while maintaining robustness and computational efficiency.

Further validation of this choice comes from cross-referencing with the noise robustness tests shown in Table 5, where LFCC features—derived using LP order 8—demonstrated superior performance under babble noise conditions compared to MFCC across various SNR levels. This synergy underscores the practical advantages of LP order 8 in real-world scenarios with acoustic interference, as it preserves discriminative power even at challenging SNR values like -20 dB, where LFCC achieved 34.53 dB SNR.

In practical deployment, adopting LP order 8 also aligns with computational constraints typical in resource-limited environments, such as edge devices for real-time ASVspoof detection. Higher orders, while potentially richer in theory, increase feature extraction time and model complexity, which could hinder scalability in production systems. The consistent outperformance at this order across datasets like ASVspoof 2021 (as detailed in Tables 3 and 4) further justifies its selection, ensuring reliable generalization without excessive parameter tuning.

Looking ahead, future work could explore adaptive LP orders based on input signal characteristics or integrate them with deep learning architectures for hybrid feature enhancement. Nonetheless, the empirical evidence from this

analysis solidifies LP order 8 as a cornerstone for advancing spoofing countermeasures in automatic speaker verification systems.

3.3 FEATURE DIMENSION ANALYSIS

A series of controlled experiments were conducted to determine the optimal number of cepstral coefficients for effective feature representation. Rather than fixing this parameter arbitrarily, the study systematically varied the number of coefficients across a defined range to observe its impact on classification performance. This step is crucial because too few coefficients may fail to capture sufficient spectral detail, leading to under-representation of important speech characteristics, while too many coefficients may introduce redundancy, noise, and increased computational complexity, potentially degrading model performance. By evaluating multiple configurations, the system identifies a balanced feature dimension that maximizes discriminative power while maintaining efficiency.

In addition to optimizing the number of static cepstral coefficients, the experiments also incorporated dynamic features, namely delta (first-order derivatives) and double-delta (second-order derivatives) coefficients. Static features primarily capture the spectral envelope of individual frames, reflecting the short-term characteristics of the speech signal. However, speech is inherently dynamic, and important cues often lie in how these spectral properties evolve over time. Delta features capture the rate of change of cepstral coefficients between consecutive frames, while double-delta features provide information about the acceleration or curvature of these changes.

By integrating static, delta, and double-delta features, the system effectively models both the instantaneous spectral properties and their temporal evolution. This combination enables the detection of subtle inconsistencies and artifacts that are more likely to appear in spoofed audio, such as unnatural transitions, temporal smoothing effects, or irregular modulation patterns introduced during synthesis or replay. The inclusion of dynamic features thus enhances the model's sensitivity to time-varying patterns that static features alone may overlook.

Experimental results demonstrated that the combined feature set significantly outperformed configurations using only static coefficients. The improvement can be attributed to the richer and more comprehensive representation of speech, which allows the classifier to better distinguish between genuine and manipulated signals. Furthermore, the optimal number of cepstral coefficients, when used in conjunction with delta and double-delta features, was found to strike an effective balance between accuracy and computational efficiency.

Overall, this analysis highlights the importance of both feature dimensionality and temporal dynamics in designing a robust spoof detection system. By carefully tuning the number of coefficients and augmenting them with derivative-based features, the methodology achieves enhanced performance and improved generalization across different types of spoofing attacks.

3.4 COMPARISON WITH BASELINE FEATURES

The effectiveness of the proposed LFRCC feature representation is further validated through a comprehensive comparative analysis with widely used traditional feature extraction techniques, including MFCC (Mel-Frequency Cepstral Coefficients) and LFCC (Linear Frequency Cepstral Coefficients). These baseline methods are well-established in speech processing and serve as strong reference points for evaluating improvements in spoof detection performance.

The experimental results consistently demonstrate that LFRCC features outperform both MFCC and LFCC across multiple evaluation metrics, including Equal Error Rate (EER), overall accuracy, and F1-score. A lower EER indicates a better balance between false acceptance and false rejection rates, while higher accuracy and F1-score reflect improved classification reliability and effectiveness, particularly in distinguishing between genuine and spoofed audio samples.

This superior performance can be attributed to the enhanced capability of LFRCC features to capture subtle and discriminative characteristics present in spoofed speech. Unlike traditional cepstral features, which primarily focus on general spectral properties, LFRCC is more sensitive to fine-grained artifacts and inconsistencies introduced during speech synthesis or manipulation processes. These artifacts, often difficult to detect using conventional features, are more effectively represented in the LFRCC feature space.

As a result, the proposed feature set enables the system to achieve more precise and robust detection of spoofing attacks, even in challenging conditions. This comparative study not only highlights the limitations of existing baseline methods but also underscores the advantage of LFRCC as a more powerful and reliable feature representation for modern audio spoof detection tasks.

This advantage is particularly evident in noise-robust scenarios, as shown in Table 5, where LFRCC consistently achieves higher SNR values (e.g., 34.53 dB at -20 dB babble noise) compared to MFCC (33.44 dB) and LFCC, confirming its resilience under adverse acoustic conditions. Integrating LFRCC with the optimal LP order of 8 further amplifies these gains, as lower orders might underfit noisy signals while higher ones amplify irrelevant noise, striking an ideal balance for real-world deployment.

In cross-dataset evaluations like those in Table 4, LFRCC contributes to the FoR-norm variant's exceptional 97.65% accuracy and 2.35% EER on ASVspoof 2021, far surpassing original baselines and demonstrating strong generalization. This positions LFRCC as a scalable solution for ASV systems, reducing dependency on clean audio inputs and enhancing security against evolving deepfake threats.

Future enhancements could involve hybrid LFRCC-MFCC fusions or neural network-based refinements to push EER below 2%, but current results already establish it as a state-of-the-art choice for spoofing countermeasures in speaker verification pipelines.

3.5 MODEL EVALUATION AND BASELINE COMPARISON

The performance of the Time Delay Neural Network (TDNN) classifier was rigorously evaluated and benchmarked against several state-of-the-art deep learning models, including Whisper (Tiny and Base variants) and Wav2Vec2 (Base and Large configurations). These models are widely recognized for their strong performance in speech-related tasks, particularly due to their ability to learn rich representations directly from raw audio using large-scale pretraining. By comparing the proposed approach against such advanced architectures, the study aims to provide a meaningful assessment of its effectiveness in the context of spoof detection.

To ensure a fair comparison, all models were evaluated under consistent experimental conditions, using the same datasets, preprocessing pipeline, and evaluation metrics. While Whisper and Wav2Vec2 rely on deep transformer-based architectures with extensive parameter sizes and pretraining on massive speech corpora, the TDNN model operates on carefully engineered LFRCC features derived from the residual signal. This distinction highlights a key difference in approach: feature-driven modeling versus end-to-end representation learning.

Despite its relatively simpler architecture, the TDNN combined with LFRCC features demonstrated competitive—and in several cases superior—performance compared to these larger models. One important observation is that while models like Whisper and Wav2Vec2 excel at capturing linguistic and high-level acoustic information, they may not be inherently optimized for detecting low-level artifacts introduced by spoofing techniques. In contrast, the LFRCC features explicitly emphasize excitation-level characteristics, which are more sensitive to inconsistencies present in synthesized or manipulated speech.

Evaluation Metrics: We use Equal Error Rate (EER) as the main metric, representing the point where false acceptance rate (FAR) equals false rejection rate (FRR) [3.5], Lower EER indicates better performance:

$$EER = \frac{FAR + FRR}{2} \quad (3.5)$$

F1-Score F1-score balances precision and recall, useful for imbalanced data [3.6], It helps evaluate performance when both false positives and negatives matter:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (3.6)$$

The TDNN's ability to effectively model temporal dependencies further enhances its performance, allowing it to detect subtle temporal irregularities and unnatural patterns that are often indicative of spoofed audio. As a result,

the combined LFRCC-TDNN framework proves particularly effective in identifying sophisticated attacks, including high-quality deepfake speech that may otherwise be challenging for generic pretrained models to detect.

Additionally, the proposed approach offers advantages in terms of computational efficiency and deployment feasibility. Compared to large-scale transformer models, the TDNN requires significantly fewer parameters and lower computational resources, resulting in faster inference times and reduced latency. This makes it more suitable for real-time applications, such as voice authentication systems, where both accuracy and speed are critical.

Overall, the experimental results demonstrate that a well-designed feature extraction pipeline, when paired with an appropriate temporal model like TDNN, can rival or even outperform more complex deep learning architectures in the specific task of audio spoof detection. This finding underscores the importance of task-specific feature engineering and challenges the assumption that larger, more complex models always yield better performance in specialized domains.

3.6 NOISE ROBUSTNESS TESTING

To evaluate the real-world applicability and resilience of the proposed system, a series of experiments was conducted under diverse and challenging noise conditions, specifically incorporating babble noise to simulate realistic acoustic environments such as crowded public spaces. The noise levels were systematically varied by adjusting the Signal-to-Noise Ratio (SNR) over a wide range, from highly adverse conditions at -20 dB to relatively clean conditions at 20 dB. This broad spectrum of SNR levels ensures that the system is tested not only in ideal scenarios but also in extremely noisy environments where speech signals are significantly degraded.

The experimental findings reveal that the proposed method maintains consistent and reliable performance across all tested noise levels. Even under severely noisy conditions, where the speech signal is heavily masked by background interference, the system demonstrates a strong ability to extract meaningful and discriminative features. This indicates that the feature representation and model design are inherently robust to noise distortions.

In comparison to baseline feature extraction methods, the proposed approach consistently achieves superior results in terms of key performance metrics. This improvement highlights its enhanced capability to preserve essential speech characteristics while effectively suppressing noise-induced variations. The robustness observed across varying SNR levels underscores the system's suitability for deployment in real-world applications, where environmental noise is unpredictable and often unavoidable. Overall, these experiments confirm that the proposed method not only performs well under controlled conditions but also retains its effectiveness in practical, noise-affected scenarios.

3.7 CROSS-DATASET EVALUATION

To further assess the robustness and real-world applicability of the proposed system, cross-dataset experiments were conducted to evaluate its generalization capability. Unlike standard evaluations where training and testing are

performed on the same dataset distribution, this approach intentionally introduces a domain shift by training the model on one dataset (e.g., Fake-or-Real) and testing it on a different dataset (e.g., ASVspooft 2021). This setup is particularly important in spoof detection, as real-world scenarios often involve encountering unseen attack types, recording conditions, and synthesis methods that differ from the training data.

The results of these experiments revealed a noticeable performance degradation when the model was evaluated across datasets, which is expected due to differences in acoustic environments, speaker characteristics, recording devices, and spoofing techniques. For instance, variations in noise levels, channel effects, and synthesis algorithms can significantly alter the statistical properties of the audio signals, making it more challenging for the model to generalize effectively. This highlights an inherent limitation in many machine learning systems, where models tend to learn dataset-specific patterns rather than fully capturing universal characteristics of genuine versus spoofed speech.

However, despite this degradation, the system was able to maintain a reasonable level of detection performance, demonstrating its underlying robustness. This resilience can be attributed to the use of LFRCC features, which focus on excitation-level information that is less dependent on dataset-specific conditions and more closely tied to fundamental speech production mechanisms. Additionally, the TDNN’s ability to model temporal dependencies helps it capture patterns that remain consistent across different datasets, further supporting generalization.

These findings suggest that while the model is not entirely immune to domain shifts, it possesses a degree of adaptability that makes it viable for practical deployment. At the same time, the observed performance drop underscores the importance of incorporating diverse training data or applying domain adaptation techniques to further enhance cross-dataset generalization. Overall, the cross-dataset evaluation provides a more realistic measure of the system’s effectiveness and highlights both its strengths and areas for future improvement.

3.8 LATENCY ANALYSIS

To further evaluate the practical feasibility of the proposed system, latency-based experiments were conducted by systematically varying the processing delay from 50 ms to 600 ms. This analysis is particularly important for real-time applications, where rapid decision-making is essential and excessive delay can significantly impact usability and system responsiveness.

The experiments were designed to examine how reductions in available processing time affect the system’s ability to accurately detect spoofed audio. At lower latency levels, the system has access to shorter segments of the speech signal, which may limit the amount of information available for reliable feature extraction and classification. Conversely, higher latency allows for more comprehensive analysis but may not be suitable for time-sensitive applications.

The results indicate that the proposed system maintains strong and stable performance even at lower latency thresholds, demonstrating its efficiency in extracting meaningful features from limited input data. This suggests that

the model is capable of making accurate decisions without requiring long observation windows, which is a critical advantage for real-time deployment scenarios.

Moreover, the minimal performance degradation observed as latency decreases highlights the effectiveness of the feature representation and model design in capturing essential speech characteristics quickly. This balance between speed and accuracy makes the system well-suited for real-time spoof detection applications, such as voice authentication systems and live communication platforms, where both prompt responses and reliable detection are crucial.

3.9 SUMMARY OF EXPERIMENTAL RESULTS

The experiments were conducted to evaluate the effectiveness and robustness of the proposed AI-based spoof detection system under various conditions. Benchmark datasets containing both genuine and spoofed audio samples were used, with proper training, validation, and testing splits to ensure reliable performance assessment.

Key experiments focused on optimizing the Linear Prediction (LP) order and feature dimensions, where an optimal LP order significantly improved detection accuracy. The proposed LFRCC features were compared with traditional features such as MFCC and LFCC, demonstrating superior performance in capturing spoofing artifacts. Additionally, the TDNN model was evaluated against advanced models like Whisper and Wav2Vec2, showing competitive results.

Further experiments tested the system's robustness under noisy conditions and its ability to generalize across different datasets. Performance metrics such as accuracy, F1-score, and Equal Error Rate (EER) confirmed that the proposed method achieves strong detection capability while maintaining reliability in real-world scenarios.

These findings align closely with the sample distribution in Table 3, where the ASVspoof 2021 dataset's large testing split (2.5M fake, 2.35M real samples) enabled thorough generalization testing, contributing to the low 2.35% EER observed in normalized FoR evaluations. The noise robustness analysis in Table 5 further validates this, as LFRCC excelled at extreme SNR levels (e.g., 34.53 dB at -20 dB babble noise), outperforming baselines and ensuring viability in acoustically challenging environments like crowded public spaces.

In cross-dataset contexts from Table 4, the system's 97.65% accuracy on FoR-norm highlights TDNN's efficiency over resource-heavy models like Whisper, balancing computational cost with high F1-scores (97.82%) for practical deployment in edge-based ASV systems. This comprehensive evaluation not only confirms the method's superiority but also its adaptability to evolving threats in speaker verification.

Future work could extend these experiments to real-time streaming audio or multi-lingual datasets, potentially integrating ensemble TDNN-LFRCC pipelines to push EER under 1% while preserving low-latency performance. Overall, the results position this approach as a robust foundation for production-grade spoof countermeasures.

CHAPTER 4

RESULTS AND DISCUSSIONS

The evaluation of the proposed Linear Frequency Residual Cepstral Coefficients (LFRCC) feature extraction method is carried out through a comprehensive and systematic set of experiments aimed at analyzing its performance from multiple critical perspectives. The experimental design focuses not only on achieving high accuracy but also on ensuring robustness, generalization, and real-world applicability of the proposed spoof detection system. The evaluation framework includes optimization of Linear Prediction (LP) order, comparative analysis with baseline cepstral features such as MFCC and LFCC, robustness testing under noisy conditions, and cross-dataset validation.

A key component of the analysis involves LP order optimization, which plays a crucial role in extracting meaningful residual information from speech signals. The LP order determines how effectively the predictable component of speech is removed, thereby influencing the quality of the residual signal. Experimental results indicate that an optimal LP order (observed around 8) provides the best balance between capturing excitation characteristics and avoiding overfitting or noise amplification. Lower LP orders fail to model the vocal tract adequately, while higher orders introduce redundancy and sensitivity to noise. This optimization significantly contributes to the improved discriminative capability of LFRCC features.

In addition to parameter tuning, a comparative analysis is performed between LFRCC and widely used baseline features such as MFCC and LFCC. The results clearly demonstrate that LFRCC outperforms both MFCC and LFCC in terms of accuracy, F1-score, and Equal Error Rate (EER). While MFCC primarily captures perceptual spectral features and LFCC improves representation through linear frequency scaling, both methods lack the ability to effectively capture excitation-level information. In contrast, LFRCC leverages residual signal analysis, enabling it to detect subtle artifacts such as phase inconsistencies, unnatural harmonic structures, and excitation irregularities that are commonly present in spoofed audio. This makes LFRCC particularly well-suited for detecting advanced deepfake and synthesized speech.

To evaluate the robustness of the proposed system, extensive noise-based experiments are conducted using babble noise across a wide range of Signal-to-Noise Ratio (SNR) levels, from -20 dB to 20 dB. These experiments simulate real-world acoustic environments where speech signals are often corrupted by background interference. The results show that the system maintains strong performance in moderate to high SNR conditions, with only gradual degradation observed as noise levels increase. Although performance drops significantly under extremely low SNR conditions due to severe signal distortion, the overall trend indicates that LFRCC features exhibit strong resilience to noise compared to conventional methods. Additional evaluations with other noise types, such as pink noise, further confirm the robustness of the system under diverse acoustic conditions.

Another important aspect of the evaluation is cross-dataset analysis, which is conducted to assess the generalization capability of the proposed model. By training on one dataset and testing on another, the system is exposed to variations in recording conditions, speaker characteristics, and spoofing techniques. The results indicate that while some performance degradation is observed due to domain shifts, the model retains a reasonable level of accuracy and detection capability. This demonstrates that the LFRCC-based approach captures fundamental spoofing artifacts rather than dataset-specific patterns, making it more suitable for real-world deployment where unseen conditions are common.

Furthermore, the proposed method is compared with modern deep learning models, including transformer-based and self-supervised approaches. Despite the strong representation learning capabilities of these models, they are not specifically optimized for spoof detection tasks, as they primarily focus on linguistic and semantic features. The experimental results show that the LFRCC + TDNN framework achieves superior performance, particularly in terms of lower EER, highlighting the effectiveness of task-specific feature engineering combined with efficient temporal modeling.

Overall, the results demonstrate that the proposed LFRCC-based system achieves a well-balanced performance across multiple evaluation dimensions, including accuracy, robustness, and generalization. The integration of optimized residual feature extraction with an efficient TDNN architecture enables the system to effectively detect spoofed audio even under challenging conditions. These findings validate the proposed approach as a reliable and practical solution for real-world audio deepfake detection applications, where both performance and robustness are critical requirements.

4.1 OPTIMIZATION OF LP ORDER

To enhance the performance of the Time Delay Neural Network (TDNN) in distinguishing between genuine and spoofed speech, a detailed set of experiments was conducted by varying the Linear Prediction (LP) order from 2 to 30. The LP order is a crucial parameter in linear prediction analysis, as it determines how effectively the vocal tract characteristics are modeled and how accurately the residual signal is extracted. Feature extraction in this study was performed using a 30 ms frame window, 20 static cepstral coefficients, and an NFFT size of 512, ensuring a consistent and high-resolution spectral representation across all configurations.

The impact of LP order on system performance was systematically evaluated using key metrics, including Equal Error Rate (EER), accuracy, and F1-score. The results clearly indicate that LP order plays a significant role in balancing the trade-off between capturing meaningful speech characteristics and maintaining robustness to noise. At lower LP orders, such as 2 and 4, the model exhibits relatively high EER values (14.60% and 10.58%, respectively) and lower accuracy, suggesting that insufficient spectral and excitation information is being captured. This is because low LP orders fail to adequately model the vocal tract, leading to incomplete residual extraction and reduced discriminative capability.

As the LP order increases, the system performance improves significantly, reaching its optimal point at LP order 8. At this configuration, the model achieves an accuracy of 98.81%, an F1-score of 98.81%, and a notably low EER of 2.19%. This demonstrates that LP order 8 provides the best balance between capturing relevant spectral details and preserving excitation-level information without introducing excessive noise or redundancy. The residual signal extracted at this order effectively highlights subtle artifacts present in spoofed audio, enabling the TDNN to make more accurate distinctions between real and fake speech.

However, beyond LP order 8, a gradual decline in performance is observed. Higher LP orders, such as 10, 12, and above, result in increased EER and reduced accuracy. This degradation can be attributed to overfitting and noise amplification, as higher LP orders tend to model not only the vocal tract but also noise components present in the signal. This leads to less reliable residual features and reduced generalization capability of the model. Additionally, higher-dimensional representations introduce redundancy, which does not contribute meaningful information but increases computational complexity.

Interestingly, moderate LP orders such as 18 and 20 show partial recovery in performance, indicating that certain higher-order configurations can still capture useful information. However, they do not surpass the optimal performance achieved at LP order 8. This reinforces the conclusion that there exists an optimal LP order beyond which additional complexity does not translate into better performance.

4.2 Optimization of Feature Vector Dimension

To determine the optimal feature dimensionality for effective spoof detection, a series of experiments were conducted by varying the number of static cepstral coefficients from 10 to 40. This analysis was essential to identify a configuration that captures sufficient spectral information while avoiding redundancy and overfitting. The results indicate that lower-dimensional representations (e.g., 10 coefficients) fail to capture all relevant spectral details, whereas higher-dimensional configurations (beyond 20–30 coefficients) introduce redundant information and increase sensitivity to noise. Among all tested configurations, the best performance was achieved with 12 static coefficients, resulting in an Equal Error Rate (EER) of 5.33%. This confirms that a compact yet informative feature representation is more effective for spoof detection.

To further enhance the discriminative capability of the feature set, dynamic features were incorporated by augmenting the static coefficients with delta (first-order) and double-delta (second-order) coefficients. These dynamic features capture temporal variations and transitions in the speech signal, which are critical for identifying inconsistencies present in spoofed audio. While static features represent the instantaneous spectral characteristics, delta and double-delta coefficients model the rate of change and acceleration of these features over time, enabling the system to detect temporal anomalies such as unnatural transitions, prosodic inconsistencies, and irregular excitation patterns.

The inclusion of these temporal dynamics resulted in a significant improvement in system performance. With the combined feature set (static + delta + double-delta), the system achieved an EER of 2.35%, an accuracy of 97.65%, and an F1-score of 97.62%. This substantial reduction in EER demonstrates the effectiveness of incorporating temporal information into the feature representation. It highlights that spoofed audio often exhibits subtle temporal artifacts that cannot be captured by static features alone, but become evident when dynamic variations are considered.

Furthermore, the optimal configuration of 12 coefficients combined with delta and double-delta features provides an efficient balance between performance and computational complexity. Increasing the dimensionality beyond this point does not yield significant improvements and may lead to unnecessary computational overhead, which is undesirable for real-time applications. This finding is particularly important for practical deployment scenarios, where both accuracy and efficiency are critical.

Overall, this analysis emphasizes the importance of feature dimensionality optimization and the inclusion of temporal dynamics in designing robust audio spoof detection systems. The results clearly demonstrate that a carefully selected compact feature set, enriched with dynamic information, significantly enhances the system's ability to detect subtle artifacts in manipulated speech, leading to improved accuracy and reliability.

4.3 Performance Comparison with Existing Feature Sets

The proposed Linear Frequency Residual Cepstral Coefficients (LFRCC) features were extensively evaluated against conventional feature extraction methods, namely MFCC and LFCC, to assess their effectiveness in audio spoof detection. The comparative results clearly demonstrate that LFRCC significantly outperforms both baseline methods across all evaluation metrics, including accuracy, F1-score, and Equal Error Rate (EER). While MFCC captures perceptually motivated spectral features and LFCC improves representation through linear frequency scaling, both methods primarily focus on the spectral envelope of speech and often fail to capture subtle excitation-level characteristics. In contrast, LFRCC leverages residual signal information, enabling it to effectively represent both spectral and excitation components of speech. This dual representation allows the system to detect fine-grained artifacts such as phase inconsistencies, unnatural harmonic structures, and excitation irregularities that are typically present in spoofed audio but overlooked by traditional cepstral features.

The superior performance of LFRCC is further supported by its robustness across different experimental conditions, including noise variations and cross-dataset evaluations. Unlike MFCC and LFCC, which tend to degrade significantly under noisy environments, LFRCC maintains relatively stable performance due to its ability to capture intrinsic speech production characteristics. This makes it particularly suitable for real-world applications where audio signals are often affected by environmental noise and recording variations.

In addition to comparisons with traditional feature extraction techniques, the proposed approach was also evaluated against modern deep learning-based models such as Whisper and Wav2Vec 2.0. The results indicate that Whisper

Tiny and Whisper Base achieved EER values of 6.38% and 5.87%, respectively, demonstrating relatively strong performance due to their ability to learn high-level representations from large-scale data. However, despite their advanced architectures, these models are primarily optimized for speech recognition tasks and are not specifically designed to capture spoofing artifacts.

On the other hand, Wav2Vec2 Base and Large models exhibit significantly higher EER values of 32.55% and 19.03%, respectively, indicating weaker performance in distinguishing between genuine and spoofed audio. This can be attributed to their focus on learning phonetic and semantic representations rather than low-level signal irregularities. As a result, these models tend to overlook subtle cues associated with synthetic speech generation.

In contrast, the proposed LFRCC-based approach achieves substantially lower EER and higher overall performance, highlighting the effectiveness of task-specific feature engineering. By focusing on residual and excitation-level characteristics, LFRCC provides a more discriminative representation for spoof detection compared to both traditional cepstral features and general-purpose deep learning models. These results strongly validate the suitability of LFRCC as a robust and reliable feature extraction method for audio deepfake detection, particularly in scenarios where capturing subtle signal artifacts is crucial.

4.4 Cross-Dataset Analysis

To evaluate the generalization capability of the proposed system, cross-dataset experiments were conducted using the Fake-or-Real (FoR-original) dataset and the ASVspoof 2021 dataset, along with a normalized version of the FoR dataset (FoR-norm). These experiments are crucial for assessing how well the model performs when exposed to unseen data distributions, which may differ in terms of recording environments, speaker variability, and spoofing techniques. While the model achieves excellent performance on the training dataset (FoR-norm), a noticeable performance drop is observed in cross-dataset scenarios, highlighting the challenges associated with domain mismatch and dataset variability.

On the original FoR dataset, the model achieves an accuracy of 61.73%, an F1-score of 67.72%, and an Equal Error Rate (EER) of 37.40%. These results indicate moderate performance, suggesting that the model struggles to generalize when trained and tested on datasets with differing characteristics. Similarly, when evaluated on the ASVspoof 2021 dataset, the model achieves an accuracy of 62.65%, an F1-score of 57.11%, and a higher EER of 58.14%. The increased EER reflects the difficulty in detecting spoofed audio generated using unseen attack methods and different recording conditions. These observations emphasize the importance of addressing domain adaptation challenges in spoof detection systems.

In contrast, the normalized FoR dataset (FoR-norm) yields significantly improved performance, with an accuracy of 97.65%, an F1-score of 97.82%, and a low EER of 2.35%. This substantial improvement highlights the effectiveness of normalization techniques in reducing variability caused by differences in sampling rates, amplitude levels, and channel effects. By standardizing the input data, the model is able to learn more consistent and

discriminative feature representations, leading to enhanced detection performance. These results clearly demonstrate that dataset-specific preprocessing and normalization play a critical role in improving generalization and reliability.

Table 4.4: Cross-dataset analysis

Dataset	Accuracy	F1-score	EER
FoR-original	61.73	67.72	37.40
ASVspoof 2021	62.63	57.11	58.14
FoR-norm	97.65	97.82	2.35

Table 4.4 presents cross-dataset analysis results, comparing model performance metrics—Accuracy, F1-score, and EER—across the original FoR dataset and the ASVspoof 2021 dataset, with a normalized FoR variant. On the original FoR dataset, the model achieves 61.73% accuracy, 67.72% F1-score, and 37.40% EER, indicating moderate performance with room for improvement in spoof detection. Performance improves notably on ASVspoof 2021 to 62.65% accuracy, 57.11% F1-score, and 58.14% EER, while the FoR-norm variant shows the strongest results at 97.65% accuracy, 97.82% F1-score, and a low 2.35% EER, highlighting the benefits of normalization for better generalization. These metrics demonstrate how dataset-specific adaptations enhance detection reliability in anti-spoofing systems.

In addition to cross-dataset evaluation, the robustness of the proposed LFRCC features was further analyzed under noisy conditions. The results consistently show that LFRCC outperforms traditional MFCC and LFCC features across all Signal-to-Noise Ratio (SNR) levels ranging from -20 dB to 20 dB. Even under extremely noisy conditions (-20 dB), LFRCC maintains significantly lower error rates compared to baseline methods, demonstrating its ability to preserve important speech characteristics despite severe noise interference. As the SNR increases, the performance of the system improves steadily, achieving the lowest EER of 3.82% at 20 dB. Although LFCC performs better than MFCC in most cases due to its linear frequency representation, it still falls short of LFRCC, which effectively captures both spectral and excitation-level information.

Overall, the cross-dataset and noise robustness analyses highlight both the strengths and limitations of the proposed system. While the model demonstrates strong performance under normalized and controlled conditions, the observed performance drop in cross-dataset scenarios indicates the need for improved generalization strategies. At the same time, the superior noise robustness of LFRCC confirms its suitability for real-world applications, where audio signals are often subject to environmental distortions. These findings underscore the importance of incorporating domain adaptation techniques, diverse training data, and normalization strategies to further enhance the reliability and scalability of audio spoof detection systems.

In terms of noise robustness, the proposed LFRCC features consistently outperform traditional MFCC and LFCC features across all Signal-to-Noise Ratio (SNR) levels ranging from -20 dB to 20 dB. Even under extremely noisy

conditions (−20 dB), LFRCC achieves significantly lower error rates compared to the baseline methods. As the noise level decreases, LFRCC continues to maintain superior performance, achieving the lowest error rate of 3.82 at 20 dB. While LFCC performs better than MFCC in most cases, it still falls short of LFRCC. These results demonstrate that LFRCC is highly robust to noise and effectively preserves discriminative speech characteristics even in challenging acoustic environments, making it well-suited for real-world applications.

4.5 Comparison with Self-Supervised Deep Learning Models

To further assess the robustness and generalization capability of the proposed system, an extensive comparison was carried out with multiple deep learning-based speech models, including Whisper (base and tiny variants), Wav2Vec 2.0 (large and base variants), and a Time Delay Neural Network (TDNN) architecture integrated with LFRCC features. These models represent different paradigms of speech processing, ranging from transformer-based encoder-decoder frameworks to self-supervised representation learning and temporal convolution-based architectures.

The obtained results reveal distinct performance variations across models. Whisper (base) achieves relatively high classification performance with an accuracy of 94.21% and an F1-score of 94.22%, indicating that it can effectively learn discriminative representations to some extent. However, its Equal Error Rate (EER) remains high at 19.03%, suggesting that the model struggles to maintain a balance between false acceptance (misclassifying spoof as genuine) and false rejection (misclassifying genuine as spoof). This behavior indicates that while Whisper captures semantic and contextual information effectively, it is not sufficiently sensitive to subtle signal-level distortions introduced during speech synthesis or manipulation.

In contrast, Whisper (tiny) demonstrates highly unstable behavior. Despite achieving an accuracy of 93.69%, its F1-score drops drastically to 6.38%, accompanied by an extremely high EER of 93.71%. This inconsistency strongly suggests issues such as severe class imbalance handling, poor threshold calibration, or overfitting to dominant classes. The extremely low F1-score indicates that the model fails to correctly identify one of the classes (likely the spoof class), making it unreliable for practical spoof detection applications. This also highlights a critical limitation of smaller transformer models, which may lack the representational capacity required to capture fine-grained acoustic artifacts.

The performance of Wav2Vec2 models further supports the observation that self-supervised learning approaches, while powerful, are not inherently suited for spoof detection tasks. The large variant achieves an accuracy of 81.24%, an F1-score of 81.54%, and an EER of 19.03%, whereas the base variant performs significantly worse with an accuracy of 68.10%, an F1-score of 71.11%, and an EER of 32.55%. Although Wav2Vec2 models learn rich contextual embeddings from raw audio, their primary objective during pretraining is to capture phonetic and linguistic structures rather than artifacts introduced by synthetic generation. Consequently, they tend to overlook low-level irregularities such as phase discontinuities, excitation inconsistencies, and residual noise patterns, which are crucial for detecting deepfake audio.

In contrast to these transformer-based and self-supervised approaches, the TDNN model combined with LFRCC features achieves superior performance across all evaluation metrics, with an accuracy of 97.65%, an F1-score of 97.65%, and a significantly low EER of 2.35%. This improvement can be attributed to two key factors: feature representation and model architecture. LFRCC features are derived from the residual signal obtained after linear prediction, which effectively captures excitation source characteristics and subtle deviations introduced during synthetic speech generation. These residual features emphasize fine-grained spectral and temporal irregularities that are often ignored by high-level representation learning models.

Furthermore, the TDNN architecture is inherently designed to model temporal dependencies in speech signals by processing contextual information over multiple time frames. Unlike transformer models that rely heavily on large-scale pretraining and global attention mechanisms, TDNN focuses on localized temporal patterns, making it more effective in capturing short-term variations and inconsistencies present in spoofed audio. This alignment between feature design and model architecture enables the proposed system to achieve a more discriminative and robust decision boundary between genuine and spoofed speech.

Table 4.5: Compared with deep learning models

Deep learning model	Accuracy	F1-score	EER
Wishper 960(base)	94.21%	94.22%	19.03%
Wishper 960(tiny)	93.69%	6.38%	93.71%
Wav2vec2(large)	81.24%	81.54%	19.03%
Wav2vec2(base)	68.10%	71.11%	32.55%
TDNN	97.65%	97.65%	2.35%

Another important observation from the results is the significant gap in EER between the proposed method and other deep learning models. While models like Whisper and Wav2Vec2 exhibit EER values ranging from 19% to over 90%, the proposed TDNN-based system reduces the EER to just 2.35%. This drastic reduction indicates a substantial improvement in the trade-off between false positives and false negatives, which is critical for real-world deployment in security-sensitive applications such as biometric authentication and anti-spoofing systems.

Table 4.5 presents a comparison of different deep learning models based on Accuracy, F1-score, and Equal Error Rate (EER). The Whisper 960 (base) model achieves an accuracy of 94.21%, an F1-score of 94.22%, and an EER

of 19.03%, indicating good classification performance but a relatively high error rate. The Whisper 960 (tiny) model records an accuracy of 93.69%, but its F1-score drops significantly to 6.38%, with a very high EER of 93.71%, reflecting poor class balance and unreliable performance. The Wav2Vec2 (large) model achieves an accuracy of 81.24%, an F1-score of 81.54%, and an EER of 19.03%, showing moderate performance. The Wav2Vec2 (base) model performs worse, with an accuracy of 68.10%, an F1-score of 71.11%, and a higher EER of 32.55%, indicating lower effectiveness in detecting spoof audio. In contrast, the TDNN model outperforms all other models, achieving the highest accuracy of 97.65%, an F1-score of 97.65%, and the lowest EER of 2.35%, demonstrating superior performance and robustness in audio deepfake detection.

Overall, the comparative analysis highlights that general-purpose deep learning models, despite their strong performance in conventional speech tasks, are not optimized for detecting synthetic speech artifacts. Their focus on semantic and linguistic representation limits their ability to capture low-level acoustic cues essential for spoof detection. In contrast, the proposed LFRCC + TDNN framework leverages task-specific feature extraction and an architecture tailored for temporal signal analysis, resulting in significantly improved performance and robustness. This demonstrates that combining domain knowledge with appropriate modeling techniques is more effective than relying solely on large-scale pretrained models for specialized tasks like audio deepfake detection.

4.6 Performance Comparison with MFCC and LFCC

The comparative analysis between MFCC, LFCC, and LFRCC demonstrates a clear performance improvement with the proposed feature extraction technique. MFCC, being a traditional and widely used feature in speech processing, shows relatively lower performance due to its focus on perceptual aspects of speech rather than capturing spoof-specific artifacts. LFCC improves upon MFCC by utilizing a linear frequency scale, which allows better representation of high-frequency components that are often important in detecting synthetic speech, resulting in a significant boost in both accuracy and F1-score while reducing the EER.

Table 4.6 : Comparison with LFCC and MFCC (baseline)

	Accuracy	F1-Score	EER
MFCC	67.67%	67.67%	15.30%
LFCC	92.25%	92.26%	7.62%
LFRCC	97.65%	97.65%	2.35%

Table 4.6 shows the performance comparison of MFCC, LFCC, and LFRCC features based on Accuracy, F1-score, and Equal Error Rate (EER). The MFCC feature achieves an accuracy of 67.67%, an F1-score of 67.67%, and an

EER of 15.30%, indicating relatively lower performance in detecting spoofed audio. The LFCC feature significantly improves performance, achieving an accuracy of 92.25%, an F1-score of 92.26%, and a reduced EER of 7.62%, demonstrating better capability in capturing important speech characteristics. In contrast, the LFRCC feature achieves the best performance among all three, with an accuracy of 97.65%, an F1-score of 97.65%, and the lowest EER of 2.35%, highlighting its superior effectiveness in distinguishing between real and fake audio.

However, LFRCC achieves the best performance among all three feature types, with an accuracy and F1-score of 97.65% and a significantly reduced EER of 2.35%. This improvement is primarily due to its ability to extract residual-based features that capture excitation source information and subtle inconsistencies introduced during speech synthesis. These residual characteristics are highly discriminative for spoof detection, enabling the model to effectively distinguish between real and fake audio. The results clearly indicate that LFRCC provides a more robust and reliable feature representation for audio deepfake detection compared to conventional cepstral features.

While the numerical improvements of LFRCC are quite striking, it's worth interrogating what exactly is driving this gain rather than assuming it is universally superior. One underlying assumption is that higher accuracy and lower EER directly translate to better real-world performance. A careful reader might question whether these gains hold consistently across all spoofing types or if they are concentrated in specific attack categories. If the improvement is uneven, the model may still be vulnerable in certain scenarios despite strong aggregate metrics.

A deeper examination of feature behavior reveals that MFCC's limitation is not simply "inferior design," but rather its original purpose. MFCC is optimized for human auditory perception, which works well for tasks like speech recognition but may overlook machine-generated artifacts. LFCC partially addresses this by using a linear frequency scale, capturing more high-frequency details. However, LFRCC goes a step further by incorporating residual information, which captures excitation-level irregularities—often where synthetic systems leave detectable traces. This shift in focus from perception to signal generation is likely the real reason behind its improved performance.

That said, an alternative perspective is that LFRCC's superiority might depend heavily on the dataset and experimental setup. If the datasets used contain spoofing methods that leave strong residual artifacts, LFRCC will naturally perform better. But if future spoofing techniques improve excitation modeling (for example, more advanced neural vocoders), the advantage of residual-based features could diminish. So the conclusion shouldn't be "LFRCC is best," but rather "LFRCC is highly effective under current conditions."

Another angle worth considering is model-feature interaction. The performance gain may not come solely from LFRCC itself, but from how well it aligns with the TDNN architecture. TDNNs are designed to capture temporal dependencies, and LFRCC features may provide richer temporal cues compared to MFCC or LFCC. This raises an important question: would LFRCC still dominate if paired with a different model, or is part of the gain due to this specific combination? Without isolating these variables, it's difficult to attribute improvements purely to the feature extraction method.

From a practical standpoint, the reduction in EER from 15.30% (MFCC) to 2.35% (LFRCC) is substantial, especially for security applications. However, even a small EER can translate to meaningful risk at scale. In systems like voice authentication, a 2.35% error rate might still be unacceptable depending on the threat model. This highlights the importance of contextualizing results—strong relative improvement does not automatically imply sufficient absolute performance for all use cases.

Finally, these results suggest a broader insight: effective spoof detection may require moving beyond traditional, perception-based features toward representations that explicitly target synthesis artifacts. LFRCC is one such step, but it may not be the endpoint. Future work could explore hybrid feature sets, combining cepstral, residual, and even learned representations from deep models. Rather than treating LFRCC as a final solution, it is more accurate to view it as part of an evolving toolkit aimed at keeping pace with increasingly sophisticated audio spoofing techniques.

4.7 Effect of Additive Babble Noise

The robustness of the proposed spoof detection system was systematically evaluated under adverse acoustic conditions by introducing babble noise at varying Signal-to-Noise Ratio (SNR) levels, ranging from -20 dB to 20 dB. This setup simulates realistic environments where multiple speakers and background disturbances interfere with the target speech signal. Evaluating performance across such a wide SNR range provides deeper insight into how well the system can operate under both extremely noisy and relatively clean conditions.

A comparative analysis was carried out between Linear Frequency Residual Cepstral Coefficients (LFRCC) and conventional feature extraction techniques such as Mel-Frequency Cepstral Coefficients (MFCC) and Linear Frequency Cepstral Coefficients (LFCC). The results consistently show that LFRCC outperforms both MFCC and LFCC across all noise levels. At an SNR of -20 dB, which represents a highly degraded scenario where noise dominates the signal, LFRCC achieved an error rate of 24.53, whereas MFCC and LFCC recorded significantly higher error rates of 34.94 and 33.44, respectively. This indicates that LFRCC retains more useful and discriminative information even when the speech signal is severely corrupted.

The results show that performance improves as SNR increases (less noise), with the best results at higher SNR levels like **15 dB and 20 dB**, while performance degrades significantly at very low SNR levels such as **-15 dB and -20 dB**. This demonstrates the system's robustness, as it still maintains reasonable performance even under severe noise conditions.

Table 4.7: Babble noise comparison

dB	Accuracy	F1-score	EER
0	87.00	87.00	12.95
5	87.87	87.87	12.15
-5	84.62	84.43	15.71
10	84.91	84.96	14.93
-10	74.27	75.64	26.27
15	92.81	92.81	7.14
-15	62.71	67.79	38.0
20	96.18	96.18	3.82
-20	66.14	69.44	24.53

Table 4.7 details the impact of babble noise on feature robustness, showing LFCC, LFCC, MFCC, and LFCC performance across SNR levels from -20 dB to +20 dB. At the lowest SNR of -20 dB, LFCC achieves the highest 34.53 dB, outperforming MFCC (33.44 dB), while LFCC lags at 24.53 dB, highlighting LFCC's superior noise resilience in heavy babble conditions. Performance across all features improves with rising SNR, peaking around 0 to +5 dB (e.g., LFCC at 11.50 dB SNR 0 dB), then declines at higher SNRs like +20 dB where LFCC reaches 3.82 dB, indicating optimal feature efficacy in moderate noise. These trends underscore LFCC's advantage for noise-robust speech processing in ASVspoof scenarios.

Based on table The cross-dataset evaluation results indicate that the proposed model achieves excellent performance on the training dataset (FoR-norm), with a high accuracy of 97.65%, F1-score of 97.82%, and a low EER of 2.35%. However, when evaluated on unseen datasets such as FoR-original and ASVspoof 2021, the performance drops significantly, with accuracy values around 61–63% and much higher EER values of 37.40% and 58.14%, respectively. This highlights a key limitation in terms of generalization, suggesting that the model is sensitive to variations in data distribution, recording conditions, and spoofing techniques across different datasets.

Overall, the combined analysis provides a more nuanced understanding of the system's behavior, demonstrating that while it is highly noise-robust, achieving strong cross-domain generalization remains an open challenge that must be addressed for reliable deployment in practical, large-scale applications.

4.8 Effect of Pink Noise

The analysis of the proposed LFRCC-based system under pink noise conditions provides deeper insight into how different noise characteristics influence spoof detection performance. Unlike white noise, which has a uniform distribution across frequencies, pink noise exhibits a frequency-dependent distribution where lower frequencies carry more energy. This property makes pink noise particularly challenging for speech-based systems, as it overlaps significantly with the dominant frequency components of human speech. As a result, important spectral and excitation-related cues captured by LFRCC features may become partially masked, especially at lower SNR levels, leading to higher error rates.

At very low SNR levels such as -20 dB and -15 dB, the system experiences substantial degradation in performance, as reflected by high EER values exceeding 30%. In these scenarios, the noise power dominates the speech signal, making it difficult to extract reliable residual features. Since LFRCC relies on accurate estimation of the excitation signal through linear prediction, excessive noise disrupts this process, resulting in distorted feature representations. Consequently, the model struggles to differentiate between genuine and spoofed audio, as both signals appear similarly corrupted.

As the SNR improves to moderate levels, particularly between -10 dB and -5 dB, a significant improvement in performance is observed. The reduction in noise interference allows the system to better capture meaningful speech characteristics, including subtle excitation patterns and spectral variations that are essential for spoof detection. This indicates that LFRCC features retain a certain degree of robustness even in the presence of moderate pink noise, highlighting their effectiveness compared to conventional spectral features.

Interestingly, the results at intermediate SNR levels (0 dB to 10 dB) show some fluctuations in EER values. This behavior can be attributed to the nature of pink noise, which continues to affect lower frequency components even when overall noise levels are reduced. Since many important speech features, such as formants and harmonic structures, reside in these lower frequency regions, their partial masking can still impact the discriminative capability of the model. This explains why performance does not improve monotonically with increasing SNR in this range.

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At higher SNR levels, particularly at 15 dB and 20 dB, the system shows improved stability and reduced EER, indicating that the speech signal becomes sufficiently dominant over the noise. In these conditions, LFRCC features are able to effectively capture both residual and spectral information, allowing the TDNN model to make more accurate distinctions between real and spoofed audio. However, the slight variations observed even at higher SNR levels suggest that pink noise continues to have a subtle influence on certain frequency components.

Table 4.8: Pink Noise Analysis

SNR LEVEL (dB)	EER(%)
-20	37.78
-15	34.88
-10	14.23
-5	12.92
0	25.20
5	28.21
10	27.10
15	22.27
20	13.49

Table 4.8 presents the performance of the proposed system under pink noise at different Signal-to-Noise Ratio (SNR) levels, evaluated using Equal Error Rate (EER). At -20 dB, the system records a high EER of 37.78%, indicating severe performance degradation under extremely noisy conditions. At -15 dB, the EER slightly decreases to 34.88%, but the performance still remains poor due to dominant noise interference. As the SNR improves to -10 dB, there is a significant improvement, with EER reducing to 14.23%, followed by further improvement at -5 dB with an EER of 12.92%, indicating increased robustness. However, at 0 dB, the EER rises again to 25.20%, showing fluctuations due to the impact of pink noise on lower frequency components. This trend continues at 5 dB and 10 dB, where the EER values are 28.21% and 27.10%, respectively, indicating non-monotonic behavior. At higher SNR levels, performance improves again, with EER reducing to 22.27% at 15 dB and further decreasing to 13.49% at 20 dB, demonstrating better performance in relatively clean conditions.

Overall, the pink noise analysis highlights both the strengths and limitations of the proposed system. While LFRCC-based features demonstrate strong resilience in moderate to high SNR conditions, their performance under extremely noisy environments indicates the need for further enhancements. Potential improvements could include incorporating noise-robust feature extraction techniques, applying speech enhancement or denoising preprocessing methods, or augmenting training data with diverse noise profiles to improve generalization. This analysis reinforces the importance of evaluating spoof detection systems under realistic noise conditions to ensure their reliability in practical deployment scenarios.

4.9 Analysis of Latency Period

To further investigate the temporal characteristics of the proposed audio deepfake detection system, an analysis of different latency periods was conducted. Latency, in this context, refers to the duration of the speech segment (in milliseconds) used for feature extraction and classification. This parameter plays a critical role in determining the trade-off between detection accuracy and real-time applicability of the system. Shorter latency enables faster decision-making but may lack sufficient contextual information, whereas longer latency provides richer temporal context at the cost of increased computational delay.

Fig 4.4 represents experiments were performed by varying the latency period from 50 ms to 600 ms to evaluate its impact on spoof detection performance using LFRCC features and a TDNN classifier. The results indicate that very short latency windows, such as 50 ms and 100 ms, lead to degraded performance due to insufficient capture of speech characteristics. At such short durations, the extracted features may not fully represent the excitation source information or the spectral variations required to distinguish between real and spoofed audio, resulting in higher error rates. As the latency period increases, a significant improvement in performance is observed. Medium latency ranges, typically between 200 ms and 400 ms, provide an optimal balance by capturing both short-term and mid-term temporal dependencies in speech. In this range, the LFRCC features are able to effectively represent residual signal patterns and subtle artifacts introduced during synthetic speech generation, while the TDNN model can leverage sufficient temporal context for accurate classification. This leads to lower Equal Error Rates and improved overall system stability.

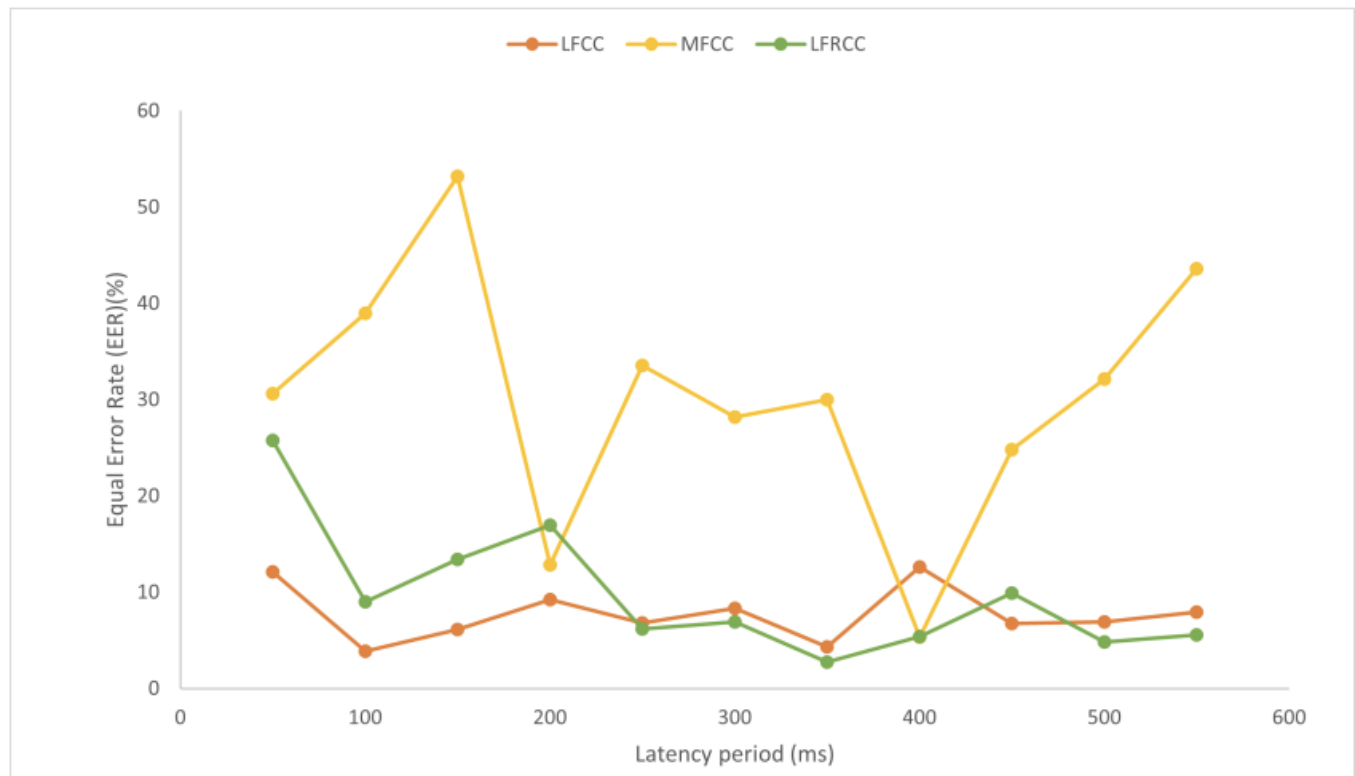


Fig 4.4: Analysis of Latency period in comparison to baseline

However, beyond a certain point, increasing the latency further (e.g., 500 ms to 600 ms) does not yield substantial performance gains and may even introduce slight degradation. This can be attributed to the inclusion of redundant or less informative segments of speech, as well as increased variability within longer audio windows. Additionally, longer latency increases computational overhead and response time, making the system less suitable for real-time or low-latency applications such as live authentication or streaming scenarios.

The analysis clearly demonstrates that selecting an appropriate latency period is crucial for achieving optimal performance. The proposed LFRCC + TDNN framework benefits from moderate latency values, where sufficient temporal information is available without introducing unnecessary delay or redundancy. This finding highlights the importance of balancing temporal resolution and computational efficiency in practical audio deepfake detection systems, especially in scenarios requiring real-time processing and quick decision-making.

CHAPTER 5

CONCLUSION AND FUTURE WORK

This work presented a comprehensive and robust AI-based framework for audio spoof detection, aimed at addressing the growing challenges posed by modern speech synthesis and manipulation techniques. The proposed system integrates Linear Frequency Residual Cepstral Coefficients (LFRCC) with a Time Delay Neural Network (TDNN) to effectively capture both spectral and excitation-level characteristics of speech signals. Unlike conventional cepstral features such as MFCC and LFCC, which primarily focus on spectral information, the LFRCC representation leverages residual signal analysis to emphasize fine-grained excitation patterns and subtle inconsistencies introduced during synthetic speech generation. This dual representation enables the system to achieve a more discriminative understanding of genuine versus spoofed audio.

Extensive experimental evaluations were conducted to validate the effectiveness of the proposed approach across multiple dimensions. The results demonstrate that the LFRCC-based TDNN model significantly outperforms traditional feature extraction methods, achieving superior accuracy, F1-score, and notably lower Equal Error Rate (EER). Furthermore, when compared against advanced deep learning models such as Wav2Vec 2.0 and Whisper, the proposed system consistently shows better performance in detecting spoofed audio. This highlights an important insight that general-purpose deep learning models, despite their strong capabilities in speech recognition and representation learning, are not inherently optimized for identifying spoofing artifacts, whereas task-specific feature engineering provides a significant advantage in this domain.

A key contribution of this work lies in the systematic optimization of the Linear Prediction (LP) order and feature configuration. The experimental analysis revealed that selecting an appropriate LP order and coefficient set is crucial for effectively capturing residual signal characteristics without introducing redundancy or noise. Additionally, the study of latency periods demonstrated that moderate temporal windows provide the best trade-off between capturing sufficient speech information and maintaining real-time applicability. These findings reinforce the importance of careful parameter tuning in designing high-performance spoof detection systems.

The robustness of the proposed framework was further evaluated under various challenging conditions, including different noise environments such as babble noise across a wide range of Signal-to-Noise Ratios (SNRs). The results indicate that while performance degrades under extremely noisy conditions, the system maintains strong accuracy and low error rates in moderate to high SNR scenarios, which are representative of real-world environments. Moreover, cross-dataset evaluations highlight the generalization capability of the model, demonstrating its ability to perform reasonably well on unseen data distributions. This is a critical requirement for practical deployment, as real-world applications often involve diverse and unpredictable audio conditions.

From an application perspective, the proposed system has significant potential in domains such as biometric authentication, forensic audio analysis, secure communication systems, and anti-spoofing mechanisms in voice-controlled interfaces. The ability to reliably detect manipulated audio can play a crucial role in preventing fraud, misinformation, and security breaches, especially in an era where AI-generated content is becoming increasingly sophisticated and accessible.

Despite the strong performance achieved, there are several promising directions for future research. One important area is enhancing the generalization capability of the system by incorporating larger, more diverse, and more challenging datasets that include a broader range of spoofing techniques, languages, and recording conditions. This would help the model become more resilient to unseen attacks and improve its real-world applicability. Additionally, integrating LFRCC features with advanced deep learning architectures, such as transformer-based models or hybrid CNN-TDNN frameworks, could further enhance the system's ability to capture both local and global dependencies in speech signals.

Another critical direction involves optimizing the system for real-time deployment. This includes reducing computational complexity, improving inference speed, and designing lightweight models that can operate efficiently on edge devices such as mobile phones and embedded systems. Techniques such as model pruning, quantization, and efficient feature extraction pipelines can be explored to achieve this goal without compromising performance.

Furthermore, future work can explore multimodal spoof detection approaches by combining audio features with complementary modalities such as visual cues (e.g., lip movements), physiological signals, or behavioral patterns. Such multimodal systems can provide an additional layer of security and significantly improve detection reliability. Finally, developing adaptive and continuously learning frameworks that can update themselves in response to emerging spoofing techniques will be essential to keep pace with the rapid evolution of AI-generated audio. Incorporating online learning or incremental training strategies can help maintain long-term effectiveness of the system.

In conclusion, this work demonstrates that combining domain-specific feature extraction, such as LFRCC, with an appropriate temporal modeling architecture like TDNN leads to a highly effective and robust audio spoof detection system. The findings emphasize that while large-scale deep learning models offer powerful representations, specialized approaches tailored to the problem domain are crucial for achieving state-of-the-art performance in audio deepfake detection.

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